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30 November 1987
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MILITARY SPECIFICATION

ANTENNA GROUP, OE-82( )/WSC-1(V)

This specification is approved for use by the Space and Naval Warfare Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements for the fabrication, testing, and production of the OE-82( )/WSC-1(V) ultra-high frequency (UHF) satellite communications (SATCOM) antenna group, hereinafter specified as the antenna group. The antenna group is for use with the AN/WSC-3 radio set for SATCOM and Line-of-Sight (LOS) communications.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-B-626	Brass, Leaded And Nonleaded, Rod, Shaped, Forgings, And Flat Products With Finished Edges (Bar And Strip)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Space and Naval Warfare Systems Command (SPAWAR 003-12), Defense Standardization Program Branch, Department of the Navy, Washington, DC 20363-5100, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 5985

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MILITARY

MIL-S-901	Shock Tests, H.I. (High-Impact), Shipboard Machinery, Equipment And Systems, Requirements For Navy
MIL-B-5423	Boots, Dust And Water Seal (For Toggle And Push-Button Switches, Circuit Breakers, And Rotary-Activated Parts), General Specification For
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-S-8516	Sealing Compound, Polysulfide Sulfur, Electrical Connectors And Electric Systems, Chemically Cured
MIL-P-15024	Plates, Tags, And Bands For Identification Of Equipment
MIL-E-15090	Enamel, Equipment, Light-Gray (Formula No. 111)
MIL-E-16400	Electronic, Interior Communication And Navigation Equipment, Naval Ship And Shore: General Specification For
MIL-E-17555	Electronic And Electrical Equipment, Accessories, Provisional Items (Repair Parts); Packaging Of
MIL-S-20708	Synchro, General Specification For
MIL-C-22751	Coating System, Epoxy-Polyamide, Chemical And Solvent Resistant; Process For Application Of
MIL-F-24044	Flanges, Coaxial Line, Rigid Air Dielectric, General Specification For
MIL-P-24441	Paint, Epoxy-Polyamide, General Specification For
MIL-C-26074	Coating, Electroless Nickel, Requirements For
MIL-T-27730	Tape, Anti-Seize, Tetrafluoroethylene, With Dispenser
MIL-C-28838	Communication Sets, UHF AN/WSC-3(V) Satellite And LOS Versions
MIL-C-28840	Connectors, Electrical, Circular-Threaded, High Density, High Shock Shipboard, Class D, General Specification For
MIL-C-39012	Connectors, Coaxial, Radio Frequency
MIL-S-46163	Sealing, Lubricating, And Wicking Compounds, Thread Locking, Anerobic Single Component
MIL-P-55110	Printed Wiring Boards

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STANDARDS

MILITARY

MIL-STD-12	Abbreviations For Use On Drawings, Specifications Standards And In Technical Documents
MIL-STD-105	Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-108	Definitions Of And Basic Requirement For Enclosure For Electric And Electronic Equipment
MIL-STD-109	Quality Assurance Terms And Definitions
MIL-STD-167-1	Mechanical Vibrations Of Shipboard Equipment (Type I-Environmental And Type II-Internally Excited)
MIL-STD-275	Printed Wiring For Electronic Equipment
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagentic Emission And Susceptibility Requirements For The Control Of Electromagnetic Interference
MIL-STD-462 and Interim Notice 5 dated 4 August 1986	Electromagnetic Interference Characteristics; Measurement Of
MIL-STD-471	Maintainability Demonstration
MIL-STD-681	Identification Coding And Application Of Hookup And Lead Wire
MIL-STD-781	Reliability Testing For Engineering Development, Qualification, And Production
MIL-STD-810	Environmental Test Methods And Engineering Guidelines
MIL-STD-965	Parts Control Program
DOD-STD-1399/300	Interface Standard For Shipboard Systems, Section 300, Electric Power, Alternating Current (Metric)
DOD-STD-1399/301	Interface Standard For Shipboard Systems, Section 301, Ship Motion and Attitude
DOD-STD-1399/302	Interface Standard For Shipboard Systems, Section 302, Weather Environment
DOD-STD-1399/702	Interface Standard For Shipboard Systems, Section 702, Synchro Data Transmission
DOD-STD-1686	Electrostatic Discharge Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)

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HANDBOOKS

MILITARY

MIL-HDBK-241	Design Guide For Electromagnetic Interference (EMI) Reduction In Power Supplies
MIL-HDBK-263	Electrostatic Discharge Control Handbook For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices) (Metric)
MIL-HDBK-781	Reliability Test Methods, Plans, And Environments For Engineering Development, Qualification, And Production

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein.

PUBLICATIONS

NAVAL AIR SYSTEMS COMMAND (NAVAIR)

WS-6536E with SCN 1 and SCN 2	Process Specification Procedures And Requirements For Preparation And Soldering Of Electrical Connections

NAVAL MATERIAL COMMAND (NAVMAT)

NAVMAT P4855-1	Navy Power Supply Reliability, Design And Manufacturing Guidelines
NAVMATINST 3401.1 (Reference (b))	Naval Material Command Nuclear Survivability Program, Nuclear Hardening Criteria

NAVAL SEA SYSTEMS COMMAND (NAVSEA)

0967-LP-597-1011	Parts Application And Reliability Information Manual For Navy Electronics Equipment

SPACE AND NAVAL WARFARE SYSTEMS COMMAND (SPAWAR)

0967-LP-614-0010	Operation And Instructions With Illustrated Parts Breakdown, Antenna Group OE-82C/WSC-1(V)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

70-1984 National Electrical Code

(Application for copies should be addressed to National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210).

Copies of specifications, standards, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer. (See 6.2)

2.3 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Antenna group. The antenna group shall be capable of maintaining upper hemispherical coverage independent of motion and maneuver of the ship. Two antennas and associated pedestals will normally be used. The two antennas will be situated such that at least one antenna is always in view of a single satellite. Hemispheric coverage from two degrees above horizon to zenith shall be required from each steerable pedestal. The two antennas, when used with a single satellite, shall track identically and simultaneously in azimuth within system tolerances from the antenna control unit. The antenna group shall require no manual inputs or operator attendance to maintain the antenna pointing to true satellite azimuth. Provision shall be made for manual azimuth positioning and control of the antennas utilizing 115 Volt (V), 400 Hertz (Hz), single-phase ship's power in lieu of the own ship heading synchro inputs. The antenna group shall permit full duplex operation. The antenna group shall be compatible with the AN/WSC-3 as specified in MIL-C-28838 without danger of preamplifier burn-out and shall allow automatic switching between LOS and SATCOM. The antenna system design shall enable reconfiguring of the control arrangement to allow simultaneous use of two separate satellites for communications. Reconfiguring to dual satellite operation shall permit simultaneous and independent control of each antenna and associated radio set. The antenna group shall conform to the requirements of MIL-E-16400, to the extent specified herein.

3.1.1 Equipment composition. The antenna group shown in FIGURE 1 is typical and for reference only. The antenna group specified herein shall consist of the units specified in a through d:

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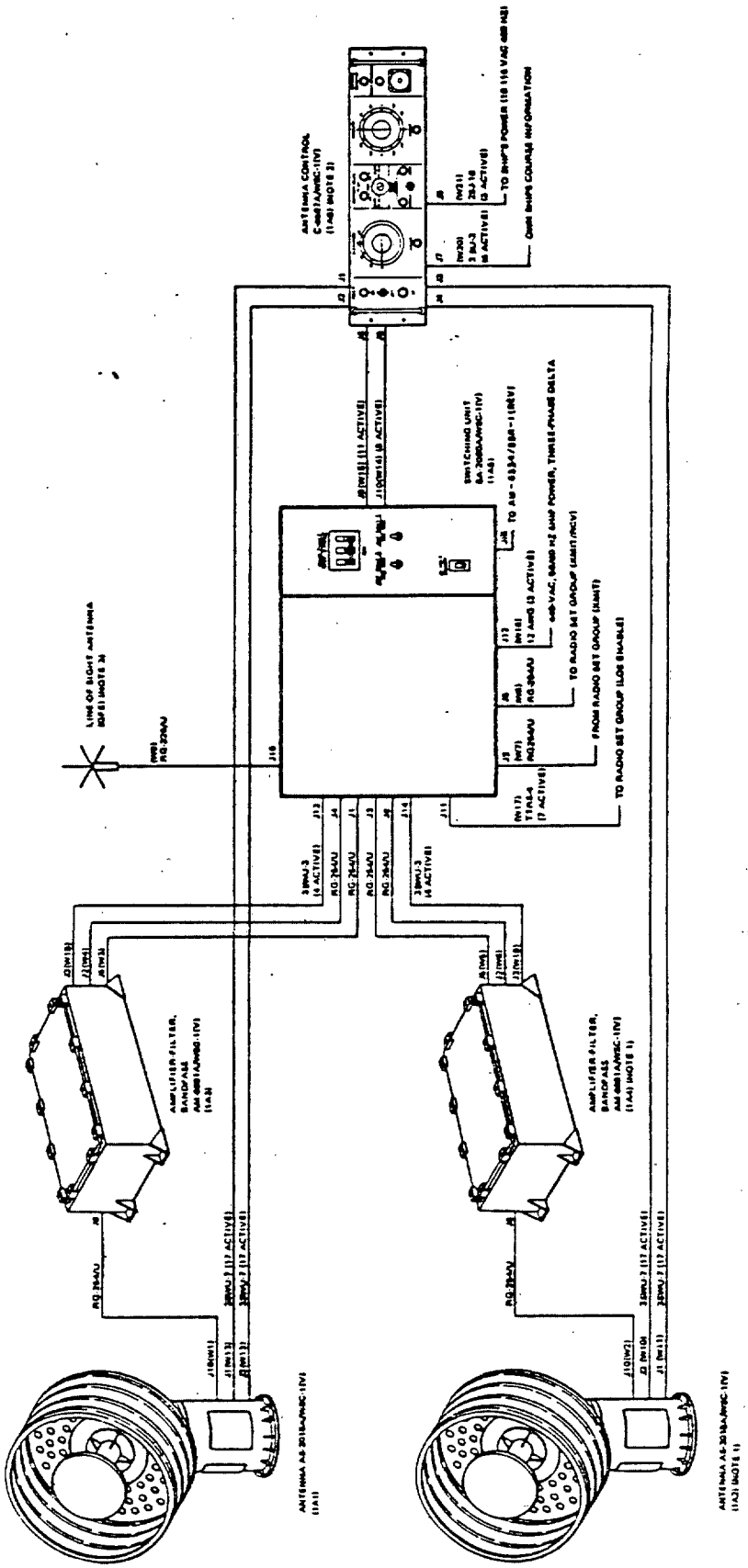


FIGURE 1. OE-82C/WSC-1(V) components.

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a. Two antenna units, AS-3018A/WSC-1(V)
b. Two amplifier-filter units, AM-6691( )/WSC-1(V)
c. One switching unit, SA-2000( )/WSC-1(V)
d. One antenna control unit, C-9597( )/WSC-1(V)

3.1.1.1 Antenna unit. The antenna shall conform to the requirements specified in 3.1.1.1.1 through 3.1.1.1.13.

3.1.1.1.1 Radiating element and reflector. The radiating element and reflector of the antenna shall conform to the requirements specified in 3.1.1.1.1.1 through 3.1.1.1.1.13.

3.1.1.1.1.1 RF Power rating. The antenna assembly shall be capable of handling 800 watts (W) peak envelope power (PEP) and 400 W average power, with no amplitude modulation (AM) of the radio frequency (RF) carrier.

3.1.1.1.1.2 Frequency range. The frequency range of operation shall be 240 megahertz (MHz) to 400 MHz.

3.1.1.1.1.3 Polarization. The polarization shall be right-hand circularly polarized for both transmit and receive.

3.1.1.1.1.4 Gain and sidelobes. The gain of the radiating assembly, when measured at the input connector on the pedestal shall be not less than 11 decibels (dB) over the frequency range 240 MHz to 400 MHz and 12 dB at 305 MHz with respect to a circularly-polarized isotropic radiator. The sidelobes shall be not less than 15 dB below the main beam maximum.

3.1.1.1.1.5 Beamwidth. The minimum 3 dB beamwidth of the antenna shall be no less than 40 degrees from 240 MHz to 271.0 MHz, no less than 33 degrees from 271.0 MHz to 311 MHz, and no less than 28 degrees from 311 MHz to 400 MHz.

3.1.1.1.1.6 Voltage standing wave ratio (VSWR). The VSWR of the radiating assembly shall be not greater than 1.5:1 across the frequency range 240 MHz to 292.2 MHz and 1.2:1 across the frequency range 292.2 MHz to 400 MHz.

3.1.1.1.1.7 Axial ratio. The axial ratio over the 240 MHz to 400 MHz frequency range shall be 3.0 dB or less within the 3 dB beamwidth.

3.1.1.1.1.8 Radiating element and reflector intermodulation. All odd ordered intermodulation products in the receive band-pass shall be at or below -125 dBm when two input signals, each at a power of 100 W, are combined and simultaneously applied to the radiating element and reflector assembly. Both unmodulated and phase shift keyed modulated carriers at data rates of up to 32,000 bits-per-second (bps) shall be used with carrier frequencies in the transmit pass-band for worst case combinations such that intermodulation products fall in the receive band-pass.

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3.1.1.1.1.9 Reflector. The back section of the reflector shall be no less than 0.203 centimeter (cm) (0.080 inch (in.)) thick aluminum. The seams shall be fully welded on the inner and outer perimeters.

3.1.1.1.1.10 Radiating element. The base of the radiating element shall be treated in accordance with MIL-C-5541, Class 3 prior to assembly. The element hybrid assembly shall be sealed with a polysulfide type sealant in accordance with MIL-S-8516.

3.1.1.1.1.11 Element mounting. All mating surfaces between the radiating element and the reflector assembly shall be treated in accordance with MIL-C-5541.

3.1.1.1.1.12 Drain holes. No exposed antenna or pedestal surfaces shall entrap water.

3.1.1.1.1.13 Reflector markings. A warning, "DO NOT PAINT WITH LEAD PAINT", shall be stencilled on the reflector using letters not less than two inches high. Instructions shall be provided on an instruction plate for repainting. These instructions shall identify the type of paint which will not degrade antenna performance.

3.1.1.1.2 Pedestal. The antenna pedestal unit shall conform to the requirements specified in 3.1.1.1.2.1 through 3.1.1.1.2.15.

3.1.1.1.2.1 Axis. Two-axis elevation over train pedestals shall be provided.

3.1.1.1.2.2 Accuracy. Each axis of the pedestal shall provide position data with an accuracy of one degree or better to the antenna control by means of synchro angle data pick-offs.

3.1.1.1.2.3 Travel. Each pedestal shall provide continuous unrestricted travel about the azimuth axis and from +2 degrees to +110 degrees in elevation, as referenced to the mounting plane.

3.1.1.1.2.4 Rotary joint. A rotary joint shall be provided in the azimuth axis. The rotary joint shall have a power handling capacity of 800 W PEP and 400 W average power. The rotary joint shall have a wow of not more than 0.1 dB per 360 degrees rotation. The insertion loss shall be 0.2 dB or less over the frequency range of 225 MHz to 400 MHz. The VSWR shall be less than 1.10:1 across the frequency range of 225 MHz to 400 MHz when terminated in 50 ohms with a VSWR less than 1.05:1. The RF connectors shall be type N female in accordance with MIL-C-39012. Rotary joint drive pins shall be QQ-B-626 brass alloy 260 or 268, hard tempered.

3.1.1.1.2.5 Slip rings. Slip rings in the train axis shall be provided for synchro information and power for the elevation drive motor.

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3.1.1.1.2.6 Angular velocity. Angular velocity shall be not less than 2.7 degrees per second in azimuth and elevation.

3.1.1.1.2.7 Main beam axis. The antenna pedestal shall position the antenna main beam axis to within five degrees of the input commands from the antenna control.

3.1.1.1.2.8 Dead band. A dead band setting of plus and minus 3.5 degrees from the null point shall be incorporated into the design to alleviate excessive stress on the azimuth drive.

3.1.1.1.2.9 Tracking. The equipments specified in 3.1.1 are not active tracking devices and do not use an RF signal originating from the satellite to detect the off-axis error and make pointing corrections. Each pedestal shall respond to the command of an operator inserted signal to the antenna control corresponding to true azimuth and elevation of the satellite. The azimuth of the antenna main beam axis shall be maintained in the correct direction after corrections have been made by the antenna control unit from the own ship heading inputs.

3.1.1.1.2.10 Telephone circuit. Circuitry for sound-power telephone communications between the pedestal and the antenna control unit shall be provided.

3.1.1.1.2.11 Sector switching. Sector switching circuitry shall be provided as an integral part of the pedestal for automatically switching from one antenna to the other. Antenna group design shall make maximum utilization of antenna pattern overlap to reduce switching occurrences. The switching zones shall be adjustable in increments of approximately seven degrees azimuth. An access opening shall be provided for use by installation personnel for sector switching adjustments during installation.

3.1.1.1.2.12 Bearing gear assembly. The bearing gear assembly shall be electroless nickel-plated in accordance with MIL-C-26074, Grade A. A grease fitting, accessible from the exterior of the pedestal, shall be installed in the pedestal housing to provide lubrication for the ball bearing path in the azimuth bearing.

3.1.1.1.2.13 Drive motor and gear box. The drive motor and gear box configuration design shall provide inherent braking and resistance to motion when drive power is removed. A brake shall not be incorporated in the design.

3.1.1.1.2.14 Pedestal grounding provision. Two grounding points shall be provided at the pedestal mounting flange, located 180 ±20 degrees apart. Each location shall be drilled and tapped for 5/16-18 UNC-2B bolt thread. A label, "GROUND HERE" shall be located next to the grounding points.

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3.1.1.1.2.15 Pedestal intermodulation. All odd ordered intermodulation products in the receive band-pass shall be at or below -125 dBm when two input signals, each at a power of 100 W, are combined and simultaneously applied to the pedestal assembly. Both unmodulated and phase shift keyed modulated carriers at data rates of up to 32,000 bps shall be used with carrier frequencies in the transmit pass-band for worst case combinations such that intermodulation products fall in the receive band-pass.

3.1.1.1.3 Weight. The weight of the antenna unit shall not be greater than 127.3 kilograms (kg) (280 pounds (lb.)).

3.1.1.1.4 Antenna unit swept volume. The radius of the antenna unit swept volume shall be not greater than 104 cm (41 in.).

3.1.1.1.5 Synchro alignment reference. The pedestal mounting flange shall have one of its 12 mounting bolt hole locations stamped with an "N" mark. This shall denote the electrical zero reference for the azimuth synchro.

3.1.1.1.6 Connector. An Electronic Industries Association (EIA) type bulkhead connector, manufacturer number 26907, part number 32255-42374, or equal shall be provided for the RF input port. The connector flange shall be in accordance with MIL-F-24044. Looking from the top down, the RF connector shall be located 80 degrees clockwise from the "N" mark.

3.1.1.1.7 RF power rating. The RF power capability of the complete antenna unit shall be 800 W PEP and 400 W average power with no amplitude modulation of the RF carrier.

3.1.1.1.8 VSWR. The VSWR measured at the input connector on the pedestal shall be not greater than 1.5:1 across the frequency range 240 MHz to 292.2 MHz and 1.2:1 across the frequency range 292.2 MHz to 400 MHz.

3.1.1.1.9 Counterbalance. Antenna-mounted subassemblies and other components of the antenna unit shall be used as counterbalances to the maximum extent possible to reduce dead weight and minimize increases in inertial loading. The smallest possible cantilever moment arms shall be used to minimize weight, inertia, and swept volume. Cable clamps shall be provided to reduce stresses on connectors, as required to withstand the shock requirement of MIL-E-16400 and the vibration requirement specified in 3.5.1.8.

3.1.1.1.10 Stops. Both electrical and mechanical stops shall be provided for the elevation axis. The mechanical stops shall operate without degradation of performance regardless of the frequency of use with the antenna being driven into the stops under full power. Means shall be provided for driving the elevation axis from the stop positions and to permit full drive power operation only in the out-of-stop direction.

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3.1.1.1.11 Antenna safety switch. Provision shall be made to disable the tracking mechanism and the RF feed of each pedestal for maintenance purposes. The switch shall be provided with a protective boot in accordance with MIL-B-5423. The other pedestal shall remain operational during a complete 360 degrees azimuth rotation when the selected pedestal is disabled. Looking from the top down, the safety switch shall be located 195 degrees clockwise from the "N" mark.

3.1.1.1.12 Thread locking. Thread locking compound as specified in MIL-S-46163, Type II, shall be applied to the threads of every screw used to mount antenna subassemblies to the pedestal subassemblies.

3.1.1.1.13 Attachment fittings. The antenna unit shall be provided with integral lifting devices. "LIFT HERE" shall be marked next to the lifting devices.

3.1.1.2 Amplifier-filter unit. Two amplifier-filter units shall be provided with each antenna group. Each amplifier-filter unit shall include a preamplifier and diplexer. The amplifier-filter unit shall conform to the requirements specified in 3.1.1.2.1 through 3.1.1.2.4.

3.1.1.2.1 Preamplifier. The preamplifier shall conform to the requirements specified in 3.1.1.2.1.1 through 3.1.1.2.1.10.

3.1.1.2.1.1 Preamplifier frequency range. The frequency range of the preamplifier shall be 243.0 MHz to 270.5 MHz.

3.1.1.2.1.2 Preamplifier bandpass characteristics. The 1 dB bandwidth shall be not less than 27.5 MHz centered about 256.75 MHz. The 3 dB bandwidth shall be 37 MHz $\pm$10 percent centered about 256.75 MHz.

3.1.1.2.1.3 Preamplifier gain. The preamplifier gain shall be 32 dB $\pm$2.0 dB at center frequency. Gain over the pass-band shall be within 1 dB of the gain at center frequency. The preamplifier gain shall be no more than 15 dB over the frequency range 292.2 MHz to 400 MHz.

3.1.1.2.1.4 Preamplifier noise figure. The noise figure over the pass-band shall not exceed 2.0 dB.

3.1.1.2.1.5 Preamplifier intermodulation. When connected to a 50-ohm resistive load with a VSWR not exceeding 2.0:1, third-order intermodulation products ($2f_2-f_1$, $2f_1-f_2$) shall be not greater than -100 decibels referenced to one milliwatt (dBm) measured at the output of the preamplifier when two -40 dBm input signals (f_1 and f_2) both unmodulated and phase-shift keyed modulated carriers at data rates of up to 32,000 bps, and located in the operating frequency range are amplified.

3.1.1.2.1.6 Preamplifier input impedance. Input impedance shall be nominally 50-ohm resistive with a VSWR not to exceed 2.0:1.

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3.1.1.2.1.7 Preamplifier output impedance. Output impedance shall be nominally 50-ohm resistive with a VSWR not to exceed 1.5:1.

3.1.1.2.1.8 Preamplifier burnout protection. Burnout protection circuitry shall be provided to protect the preamplifier and the remainder of the receiver chain. The frequency range of the burnout protection shall be from 210 MHz to 275.5 MHz. The burnout protection shall possess the properties specified in a through e below:

a. Interfering signals up to +35 dBm shall be limited to a level which shall not cause the receiver performance (signal to noise ratio) to be degraded by more than 0.2 dB.

b. Insertion loss at normal signal level shall be not greater than 0.4 dB.

c. Insertion loss at high signal levels shall be great enough to prevent damage to the receive modules. The insertion loss shall be a minimum of 20.0 dB.

d. Input and output impedances of the burnout protectors shall be a nominal 50 ohms with a VSWR not to exceed 1.5:1.

e. A bandwidth-shaped amplifier is required to provide isolation between transmit and receive modes.

3.1.1.2.1.9 Interior coating. Interior areas affected by dry-air pressurization will be treated in accordance with MIL-C-5541, Class 3.

3.1.1.2.1.10 Pressurization fitting. The preamplifier shall have an air fitting to permit dry-air pressurization.

3.1.1.2.2 Diplexer. The diplexer shall conform to the requirements specified in 3.1.1.2.2.1 through 3.1.1.2.2.10.

3.1.1.2.2.1 Diplexer transmit channel insertion loss. The insertion loss from the transmit port to the antenna port shall not exceed 0.85 dB over the frequency range of 292.2 MHz to 400 MHz.

3.1.1.2.2.2 Diplexer transmit channel ripple. The insertion loss ripple of the transmit channel shall be not greater than ±0.50 dB over the transmit band of 292.2 MHz to 400 MHz.

3.1.1.2.2.3 Diplexer receive channel insertion loss. The insertion loss from the antenna port to the receive port shall not exceed 1.0 dB over the frequency range of 243.0 MHz to 270.5 MHz.

3.1.1.2.2.4 Diplexer receive channel ripple. The insertion loss ripple of the receive channel shall be not greater than ±0.50 dB over the receive band of 243.0 MHz to 270.5 MHz.

3.1.1.2.2.5 Diplexer transmit-to-receive channel isolation. The receive channel shall provide at least 60 dB attenuation for all frequencies from direct current (DC) to 205 MHz. The isolation from the diplexer common port to the receiver port shall be not less than 95 dB from 205 MHz to 225 MHz. The filter attenuation shall be as high as practical from 225 MHz to 243 MHz,

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but consistent with the insertion loss of the receive pass-band. Similarly, from 270.5 MHz to 285 MHz the filter attenuation shall be as high as practical, but consistent with the insertion loss of the receive band. From 285 MHz to 292.2 MHz, the attenuation shall be not less than 60 dB. For the transmit pass-band (292.2 MHz to 400 MHz) the attenuation shall be not less than 100 dB. An attenuation of not less than 60 dB shall be required from 400 MHz to 1 gigahertz (GHz) and not less than 40 dB between 1 GHz and 10 GHz, except between 2 GHz and 3 GHz where the attenuation shall be not less than 20 dB. Coupling from the transmit to receive channels shall be attenuated not less than 100 dB in the transmit band and shall be not less than 90 dB for any frequency in the receive band. The transmit channel shall provide a minimum of 90 dB attenuation for the frequencies between 243.0 MHz and 270.5 MHz, and 30 dB attenuation for third harmonics from 876.6 MHz to 1200 MHz.

3.1.1.2.2.6 Diplexer VSWR. The VSWR of the receive ports shall be less than 1.5:1 in their frequency bands for a 50-ohm impedance line with remaining ports matched. The VSWR of the transmit ports shall be less than 1.2:1 in their frequency bands for a 50-ohm impedance line with remaining ports matched.

3.1.1.2.2.7 Diplexer power capability. The diplexer shall be capable of operating with 800 W PEP and 400 W average power.

3.1.1.2.2.8 Diplexer intermodulation. When connected to a 50-ohm resistive load with a VSWR not exceeding 1.2:1, all odd-order intermodulation products shall be at or below -115 dBm when two input signals, each with a power of 100 W, are combined and simultaneously applied to the transmit path of the diplexer. Both unmodulated and phase-shift-keyed modulated carriers at data rates of up to 32,000 bps shall be used with carrier frequencies in the transmit pass-band selected for worst case combinations such that intermodulation products fall in the receive bandpass. At no time shall intermodulation products be detected within the receiver bandpass higher in level than specified in 3.1.1.2.2.5.

3.1.1.2.2.9 Interior coating. Interior areas affected by dry-air pressurization shall be treated in accordance with MIL-C-5541, Class 3.

3.1.1.2.2.10 Pressurization fittings. The diplexer shall have an air fitting to permit dry-air pressurization and an air fitting to permit purging.

3.1.1.2.3 Bonding and grounding. Bonding and grounding location and requirements shall be as shown in the Amplifier-Filter, Bandpass 1A3 and 1A4, Installation Drawing figure of NAVELEX 0967-614-0010.

3.1.1.2.4 Weight. The weight of the amplifier-filter unit shall not exceed 47.7 kg (105 lb.).

3.1.1.3 Switching unit. The switching unit shall conform to the requirements specified in 3.1.1.3.1 through 3.1.1.3.13.

3.1.1.3.1 SATCOM antenna transmit-transfer relay. The transmit-transfer relay shall be an integral part of the switching unit and shall be used for automatically switching from one antenna to the other. The transmit-transfer relay shall conform to the requirements specified in 3.1.1.3.1.1 through 3.1.1.3.1.7.

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3.1.1.3.1.1 Function. The transmit-transfer relay shall connect the desired antenna subsystem to the transmitter. It shall perform a make-before-break function in order that an open circuit is never presented to the transmitter. The relay shall provide an auxiliary set of contacts which are used to supply 28 volts direct current (VDC) to the IN USE indicators in the antenna control unit when the relay is energized for the selected antenna.

3.1.1.3.1.2 Switching time. Double termination of the transmission line shall not exist for longer than 10 milliseconds (ms) during the make-before-break interval provided by the antenna control.

3.1.1.3.1.3 Power. The contacts shall be capable of handling 800 W PEP and 400 W average power.

3.1.1.3.1.4 VSWR. The VSWR of all ports shall not exceed 1.15:1 when unused ports are terminated in 50 ohms with a VSWR of less than 1.05:1.

3.1.1.3.1.5 Insertion loss. The insertion loss shall not exceed 0.2 dB.

3.1.1.3.1.6 Isolation between ports. Isolation from port-to-port shall be not less than 50 dB.

3.1.1.3.1.7 Operating voltage. The transmit-transfer relay shall operate from the 28 VDC power supply.

3.1.1.3.2 Receiver-transfer relay. The receiver-transfer relay shall be an integral part of the switching unit and shall be used for automatically switching from one antenna to the other. The receiver-transfer relay shall conform to the requirements specified in 3.1.1.3.2.1 through 3.1.1.3.2.5.

3.1.1.3.2.1 Switching time. The switching time shall not exceed 25 ms.

3.1.1.3.2.2 Insertion loss. The insertion loss shall not exceed 0.2 dB.

3.1.1.3.2.3 VSWR. The VSWR of all ports shall not exceed 1.2:1 when unused ports are terminated in 50 ohms with a VSWR of less than 1.05:1.

3.1.1.3.2.4 Power. The contacts shall be capable of handling 800 W PEP and 400 W average power.

3.1.1.3.2.5 Operating voltage. The receiver-transfer relay shall operate from the 28 VDC power supply.

3.1.1.3.3 SATCOM-LOS relays. The SATCOM-LOS relays shall be integral parts of the switching unit. One relay shall switch the AN/WSC-3 transmit line between the SATCOM and LOS antennas. The second relay shall switch the AN/WSC-3 receive line between the SATCOM and LOS antennas. The SATCOM-LOS relays shall conform to the requirements specified in 3.1.1.3.3.1 through 3.3.1.3.3.6.

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3.1.1.3.3.1 Isolation between ports. The isolation between ports shall be not less than 60 dB.

3.1.1.3.3.2 Switching time. The switching time shall not exceed 25 ms.

3.1.1.3.3.3 Insertion loss. The insertion loss shall not exceed 0.2 dB.

3.1.1.3.3.4 VSWR. The VSWR of all ports shall not exceed 1.2:1 when unused ports are terminated in 50 ohms with a VSWR of less than 1.05:1.

3.1.1.3.3.5 Power. The contacts shall be capable of handling 800 W PEP and 400 W average power.

3.1.1.3.3.6 Operating voltage. The SATCOM-LOS relays shall operate from the 28 VDC power supply.

3.1.1.3.4 LOS transmit-receive relay. The LOS transmit-receive relay shall be an integral part of the switching unit to switch the LOS antenna between the SATCOM-LOS relays specified in 3.1.1.3.3 and shall be as specified in 3.1.1.3.4.1 through 3.1.1.3.4.7.

3.1.1.3.4.1 Isolation between ports. The isolation between ports shall be not less than 60 dB.

3.1.1.3.4.2 Switching time. The switching time shall not exceed 25 ms.

3.1.1.3.4.3 Insertion loss. The insertion loss shall not exceed 0.2 dB.

3.1.1.3.4.4 VSWR. The VSWR of all ports shall not exceed 1.2:1 when unused ports are terminated in 50 ohms with a VSWR of less than 1.05:1.

3.1.1.3.4.5 Power. The contacts shall be capable of handling 800 W PEP and 400 W average power.

3.1.1.3.4.6 Operating voltage. The LOS transmit-receive relay shall operate from the 28 VDC power supply.

3.1.1.3.4.7 Relay control. The relay shall operate from open circuit or ground input commands. Ground shall enable the relay.

3.1.1.3.5 Keyline interlock control. A connector (J11 on "DC Circuits, Maintenance Interconnection Diagram" figure of NAVELEX 0967-LP-614-0010) shall provide the interfaces specified in a through c:

a. Selection of LOS mode - a switch closure on the front of the radio shall switch the antenna RF system from the SATCOM antenna to the LOS antenna.

b. Selection of LOS transmit mode - shall enable the radio to transmit through the LOS antenna when the keyline is enabled. When the keyline is not enabled, the radio shall be in receive.

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c. Transmitter keyline interlock - transfers the keyline interlock function from the SATCOM antennas to the LOS antenna.

The pin-out of this connector is specified in 3.1.1.3.5.1 and 3.1.1.3.5.2.

3.1.1.3.5.1 Control for single satellite coverage. Control for single satellite coverage shall be as specified in a through g:

a. Pin A. Keyline interlock (The switching unit shall provide an internal closure between pins A and F and pins B and E when in the LOS mode. Also, in the SATCOM mode, the switching unit shall provide an internal closure between pins A and B when the active SATCOM antenna is enabled for transmit. Pins A and B shall be open when the antenna safety (POWER ON/SAFE) switch of the active SATCOM antenna is in the SAFE position.)
b. Pin B. Keyline interlock (see a.)
c. Pin C. 28 VDC power supply RETURN
d. Pin D. LOS enable (external closure between pins C and D activates the relays specified in 3.1.1.3.3 to the LOS mode)
e. Pin E. LOS antenna interlock (see a.)
f. Pin F. LOS antenna interlock (see a.)
g. Pin G. Transmit (external closure between pins C and G activates the relay specified in 3.1.1.3.4 to the LOS transmit mode)

3.1.1.3.5.2 Control for dual satellite coverage. Control for dual satellite coverage shall be as specified in a through d:

a. Pin A. Keyline interlock (The switching unit shall provide an internal closure between pins A and B when antenna no. 1 (as defined in TABLE I and TABLE II) is enabled for transmit. Pins A and B shall be open when the safety (POWER ON/SAFE) switch of antenna no. 1 is in the SAFE position.)
b. Pin B. Keyline interlock (see a.)
c. Pin H. Keyline interlock (This adds a second keyline interlock for antenna no. 2 in the dual satellite mode. The switching unit shall provide an internal closure betwen pins H and J when antenna no. 2 is enabled for transmit. Pins H and J shall be open when the antenna safety (POWER ON/SAFE) switch of antenna no. 2 is in the SAFE position.)
d. Pin J. Keyline interlock (see c.)

3.1.1.3.6 28 VDC power supply. A 28 ±4 VDC power supply shall be provided as an integral part of the switching unit to provide power for coaxial relays and the preamplifier.

3.1.1.3.7 Stepdown transformer. A stepdown transformer shall be provided as an integral part of the switching unit to accept shipboard primary power and transform to the required voltage to power the respective components. The stepdown transformer shall conform to the requirements specified in 3.1.1.3.7.1 and 3.1.1.3.7.2.

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3.1.1.3.7.1 Input voltage. The input voltage shall be 440 V, three phase delta, in accordance with DoD-STD-1399, Section 300, Type I except the the nominal frequency shall be 50 Hz to 60 Hz.

3.1.1.3.7.2 Output voltage. The output voltage shall be 208 V, three phase wye, 50 Hz to 60 Hz.

3.1.1.3.8 Auxiliary receive output port. An auxiliary receive output port shall be provided for use with an additional receiver. The port shall be connected to an internal receive signal tap from a directional coupler with a fixed non-adjustable signal level no more than a nominal 20 dB below the signal received from each antenna. In the dual satellite mode, received signals from each antenna shall be sampled and combined in an RF hybrid. In this instance, the signal level at the auxiliary receive output port shall be no more than 4.0 dB below the output of the internal receive signal tap.

3.1.1.3.9 Switching unit intermodulation. All odd ordered intermodulation products in the receive band-pass shall be at or below -80 dBm when two input signals, each at a power of 100 W are combined and simultaneously applied to the transmit path only of the switching unit assembly. Both unmodulated and phase shift keyed modulated carriers at data rates of up to 32,000 bps shall be used with carrier frequencies in the transmit pass-band for worst case combinations such that intermodulation products fall in the receive band-pass.

3.1.1.3.10 Air pressure switch. The air pressure switch shall provide a set of contacts which shall close when input dry-air pressure is 1.5 pounds per square inch, gauge (psig) or less. The contacts shall be rated at 1 ampere, 28 VDC. The dry-air input shall be in accordance with 3.1.1.5.1.5.

3.1.1.3.11 Pressure regulation system. A pressure-regulated system shall be incorporated for use with the dry air system. The regulated system shall limit the pressure for pressurized subassemblies to 3 psig $\pm$ 1 psig. In the event of the loss of input air pressure, the system shall not lose more than 0.5 psig pressure in four hours. The pressure regulation system shall exhaust all input pressures above 4 psig up to 150 psig and shall vent the air to the exterior of the antenna system.

3.1.1.3.12 Weight. The weight of the switching unit shall not exceed 54.5 kg (120 lb.).

3.1.1.3.13 Satellite coverage switch. The switching unit shall have a switch designated S1. In position 1, the switch shall configure the switching unit for single satellite coverage. In position 2, the switch shall configure the switching unit for dual satellite coverage by reconfiguring J11 to eliminate the LOS capability and add the separate keyline interlock for antenna no. 2.

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3.1.1.4 Antenna control unit. An antenna control unit, configured for 19 inch cabinet installation, shall be provided for control of the antenna group. The antenna control unit shall conform to the requirements specified in 3.1.1.4.1 through 3.1.1.4.7.

3.1.1.4.1 Antenna control function. The antenna control unit shall include antenna pedestal control necessary to accomplish the subsystem functions, azimuth slewing controls, and an automatic antenna sector switching function. It shall control either or both pedestals. The antenna control unit shall process operator input commands of true elevation and azimuth and own ship heading inputs from the own ship data set and convert them to commands for antenna pointing. Relative azimuth positioning and elevation positioning shall be provided by using the ship's 115 V, 400 Hz single-phase power in lieu of own ship heading inputs. The antenna control shall provide for slewing manually in azimuth both antennas independent of the own ship heading inputs and independent of each other.

3.1.1.4.2 Form and fit. The antenna control unit shall have the physical dimensions of FIGURE 2.

3.1.1.4.3 Controls and indicators. Controls, indicators, and functions shall be provided as indicated in FIGURE 3, TABLE I, and TABLE II.

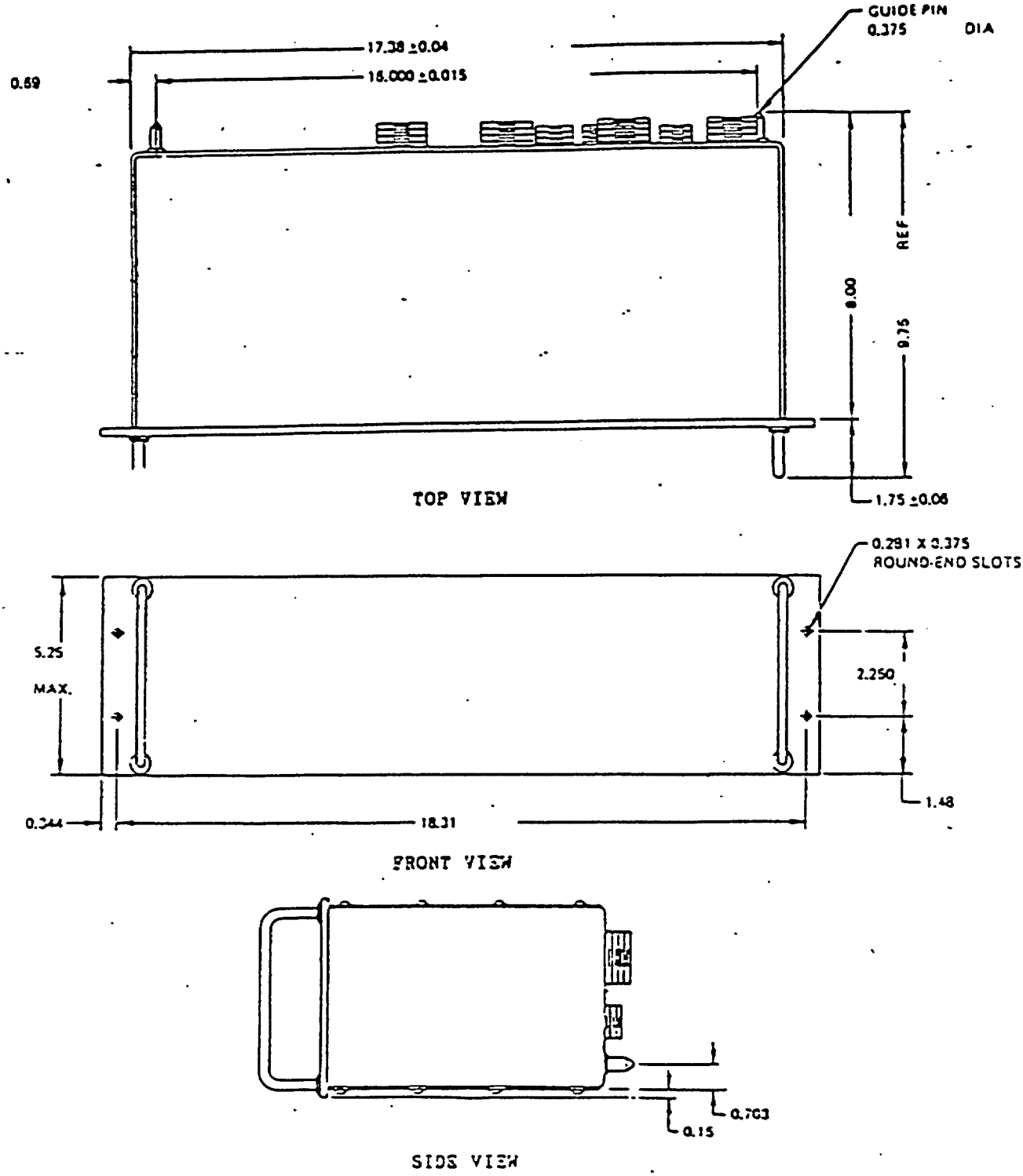
3.1.1.4.4 Manual override. Provision shall be made for manually overriding the antenna blocked sector switch and selecting one of the antennas for operation. Provisions shall be made to automatically override the sector switch blockage of one antenna when the other antenna is disabled. This override control shall be mounted on the front of the antenna control unit and be clearly labeled.

3.1.1.4.5 Antenna control configurations. The antenna control shall have two operating configurations as specified in 3.1.1.4.5.1 and 3.1.1.4.5.2. The desired configuration shall be selectable by a two-position switch (S4) located inside the antenna control unit. The switch shall be labeled SATELLITE CONFIGURATION and the two positions shall be marked with SINGLE and DUAL. The switch shall be accessible only by removing the cover of the antenna control unit. The selected configurations shall be compatible with the switching unit which has the corresponding configurations selectable by a two-position satellite configuration switch with corresponding SINGLE and DUAL positions. Switch S1 shall also be accessible only by removing the cover of the switching unit.

3.1.1.4.5.1 Configuration 1 - single satellite coverage. In configuration 1, both antennas are pointed towards the same satellite. When one antenna enters a blocked sector (dictated by blockage caused by ship's structure or radiation hazard), transmit and receive capabilities are automatically transferred to the other antenna.

3.1.1.4.5.2 Configuration 2 - dual satellite coverage. In configuration 2, the two antennas are pointed towards different satellites. No antenna handover (transfer of transmit and receive capabilities) shall occur in this configuration. However, if an antenna enters a blocked sector, transmit by that antenna shall be inhibited by virtue of the corresponding keyline interlock interface at the switching unit.

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NOTE: Dimensions are in inches.

FIGURE 2. Antenna control, C-9597( )/WSC-1(V), dimensions.

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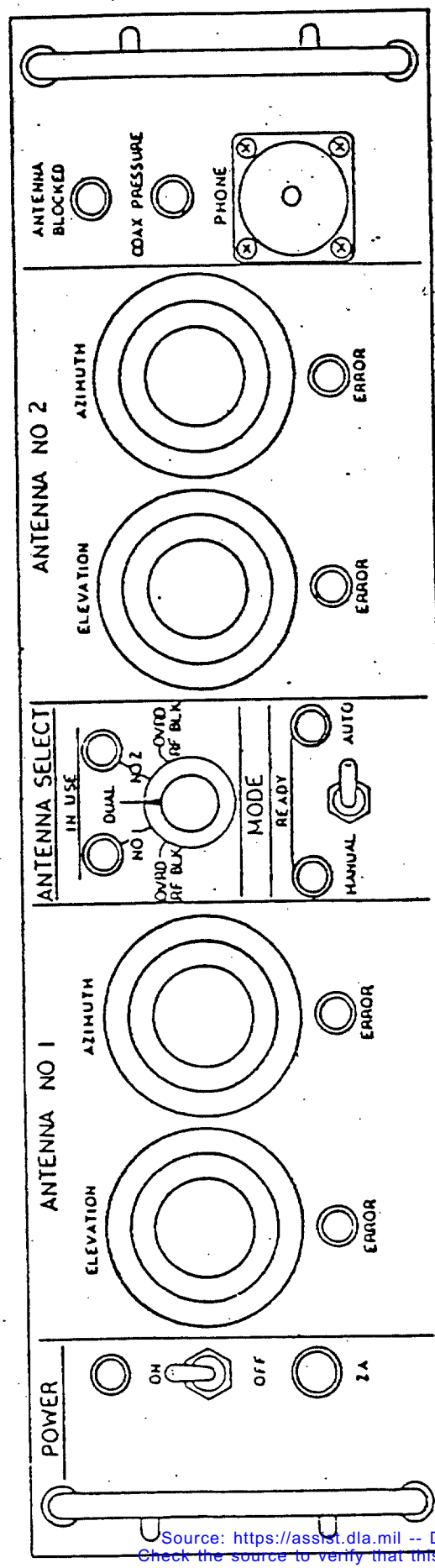


FIGURE 3. Antenna control unit, C-9597( )/WSC-1(V), front panel layout.

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TABLE I. Antenna control C-9597( )/WSC-1(V), controls and indicators - single satellite coverage configuration.

NOTE: Applicable only when the antenna group is configured for single satellite, dual antenna with options selected as specified in a and b:

a. Satellite configuration switch, (S4), in antenna control unit is in position "1".
b. Satellite configuration switch, (S1), in switching unit is in position "1".
c. The switching unit RF cables are connected for single satellite operation.

Control or indicator marking	Type	Function
POWER ON-OFF	Toggle switch	Controls 28 VDC power and 115 VAC, 400 Hz synchro reference power to antenna control and pedestals
POWER ON	Indicator (green)	When lit, indicates DC power is applied to antenna control and pedestal and 115 volts alternating current (VAC), 400 Hz ship's power and Own Ship Heading inputs are energized, provided the two READY lamps (MANUAL and AUTO) are also lit
ANTENNA NO 1, ELEVATION	Control knob	Selects commanded antenna (no. 1) angle in the elevation plane between +2 degrees and +110 degrees relative to the antenna mounting base
ANTENNA NO 1, AZIMUTH	Control knob	Selects commanded antenna (no. 1) angle in the azimuth plane relative to true north (when MODE switch is at AUTO), or relative to ship heading (when MODE switch is at MANUAL)
ANTENNA SELECT, IN USE (NO 1)	Indicator (white)	When lit, indicates that antenna no. 1 is connected to the radio set group
OVRD RF BLOCK, NO 1, DUAL, NO 2, OVRD RF BLOCK	Rotary switch	This switch has 5 positions. Permits selection of either or both antennas (in a two-antenna installation), or the connected antenna (in a one-antenna installation) with the radio set group. In DUAL position, the antenna not blocked is automatically selected

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TABLE I. Antenna control C-9597( )/WSC-1(V), controls and indicators - single satellite coverage configuration. - (Continued)

Control or indicator marking	Type	Function
		for use. If neither is blocked, no. 1 antenna is preferred and automatically selected unless it had previously been blocked. If both antennas are blocked, the ANTENNA BLOCKED lamp is lit (see ANTENNA BLOCKED). In NO 1 (or NO 2) position, antenna no. 1 (or no. 2) is manually selected and power is removed from the control circuits of the alternate antenna. In OVRD BLK position NO 1 (or NO 2), contacts of the ANTENNA SELECT switch bridge the sector switch contacts in the antenna pedestal to maintain the selected antenna operative even though the blocked sector is entered
ANTENNA SELECT, IN USE (NO 2)	Indicator (white)	When lit, indicates that antenna no. 2 is connected to the radio set group
ANTENNA NO 2, ELEVATION	Control knob	Selects commanded antenna (no. 2) angle in the elevation plane between +2 degrees and +110 degrees relative to the antenna mounting base
ANTENNA NO 2, AZIMUTH	Control knob	Selects commanded antenna (no. 2) angle in the azimuth plane relative to true north (when MODE switch is at AUTO), or relative to ship heading (when MODE switch is at MANUAL)
COAX PRESSURE	Indicator (red)	When lit, indicates that the pressure in the dry air system is below 1.5 psig
ANTENNA BLOCKED	Indicator (red)	When lit, indicates that both antennas of a two-antenna installation are blocked or that the single antenna of a one-antenna installation is blocked. (If one antenna of a two-antenna installation is shut down

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TABLE I. Antenna control C-9597( )/WSC-1(V), controls and indicators - single satellite coverage configuration. - (Continued)

Control or indicator marking	Type	Function
		and the ANTENNA SELECT switch is at OVRD RF BLK position for the other antenna, the ANTENNA BLOCKED lamp will not light when the selected antenna is in the blocked sector)
PHONE	Telephone jack	Provides access to sound-powered intercom circuit with both antenna pedestals
ANTENNA NO 2, AZIMUTH ERROR	Indicator (red)	When lit, indicates a discrepancy between the AZIMUTH command angle and the actual antenna angle. Lamp is extinguished after antenna moves to the commanded azimuth angle
ANTENNA NO 2, ELEVATION, ERROR	Indicator (red)	When lit, indicates a discrepancy between the ELEVATION command angle and the actual antenna angle. Lamp is extinguished after antenna moves to the commanded elevation angle
MODE READY, AUTO	Indicator (amber)	When lit, indicates that Own Ship Heading inputs are present at the antenna control unit
MANUAL, AUTO	Toggle switch	In AUTO position, Own Ship Heading inputs are used as a reference to provide true azimuth command signals during ship maneuvers. In MANUAL position, the ship's 115 VAC, 400 Hz power is used to provide a reference for azimuth command signals that is only relative to the heading of the ship
READY, MANUAL	Indicator (amber)	When lit, indicates that the ship's 115 VAC, 400 Hz power is available at the antenna control unit

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TABLE I. Antenna control C-9597( )/WSC-1(V), controls and indicators - single satellite coverage configuration. - (Continued)

Control or indicator marking	Type	Function
ANTENNA NO 1, AZIMUTH, ERROR	Indicator (red)	When lit, indicates a discrepancy between the AZIMUTH command angle and the actual antenna angle. Lamp is extinguished after antenna moves to the commanded azimuth angle
ANTENNA NO 1, ELEVATION ERROR	Indicator (red)	When lit, indicates a discrepancy between the ELEVATION command angle and the actual antenna angle. Lamp is extinguished after antenna moves to the commanded elevation angle
2A	Fuse holder	Contains 2-ampere fuse that protects the 28 VDC input circuits of the antenna control unit
S4	Rotary switch (located inside antenna control unit and accessible only by removing cover)	In position 1, configures antenna control unit for single satellite coverage In position 2, configures antenna control unit for dual satellite coverage by making antenna no. 2 separately controlled

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TABLE II. Antenna control C-9597( )/WSC-1(V), controls and indicators - dual satellite coverage configuration.

NOTE: Applicable only when the antenna group is configured for dual satellite, single antenna with options selected as specified in a and b:

a. Satellite configuration switch, (S4), in antenna control unit is in position "2".
b. Satellite configuration switch, (S1), in switching unit is in position "2".
c. The switching unit RF cables are connected for dual satellite operation.
d. All antenna sector cams have been set for "no block" on both antennas.

Control or indicator marking	Type	Function
POWER ON-OFF	Toggle switch	Controls 28 VDC power and 115 VAC, 400 Hz synchro reference power to antenna control and pedestals
POWER ON	Indicator (green)	When lit, indicates DC power is applied to antenna control and pedestal(s) and 115 VAC, 400 Hz ship's power and Own Ships Heading inputs are energized, provided the two READY lamps (MANUAL and AUTO) are also lit
ANTENNA NO 1, ELEVATION	Control knob	Selects commanded antenna (no. 1) angle in the elevation plane between +2 degrees and +110 degrees relative to the antenna mounting base
ANTENNA NO 1, AZIMUTH	Control knob	Selects commanded antenna (no. 1) angle in the azimuth plane relative to true north (when MODE switch is at AUTO), or relative to ship heading (when MODE switch is at MANUAL)
ANTENNA SELECT, IN USE (NO 1)	Indicator (white)	Always lit in dual satellite operation except when antenna safety switch is in SAFE position. Indicates antenna off when indicator is out and ANTENNA BLOCKED lamp is illuminated.

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TABLE II. Antenna control C-9597( )/WSC-1(V), controls and indicators - dual satellite coverage configuration. - (Continued)

Control or indicator marking	Type	Function
OVRD RF BLOCK, NO 1, DUAL, NO 2, OVRD RF BLOCK	Rotary switch	This switch has 5 positions but should remain in DUAL position for two-satellite operation. In dual satellite operation, both antenna IN USE indicators will be illuminated, indicating dual satellite operation and drive power is available for both antennas In NO 1 - OVRD RF BLOCK, the switch overrides the RF blocked sector of antenna no. 1 In NO 2 - OVRD RF BLOCK, the switch overrides the RF blocked sector of antenna no. 2
ANTENNA SELECT, IN USE (NO 2)	Indicator (white)	Always lit in dual satellite operation except when antenna safety switch is in SAFE position. Indicates antenna off when indicator is out and ANTENNA BLOCKED lamp is illuminated.
ANTENNA NO 2, ELEVATION	Control knob	Selects commanded antenna (no. 2) angle in the elevation plane between +2 degrees and +110 degrees relative to the antenna mounting base
ANTENNA NO 2, AZIMUTH	Control knob	Selects commanded antenna (no. 2) angle in the azimuth plane relative to true north (when MODE switch is at AUTO), or relative to ship's heading (when MODE switch is at MANUAL)
COAX PRESSURE	Indicator (red)	When lit, indicates that the pressure in the dry air system is below 1.5 psig
ANTENNA BLOCKED	Indicator (red)	When lit, indicates that an antenna is in a blocked sector. The IN USE indicators permit determination of antenna that is blocked

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TABLE II. Antenna control C-9597( )/WSC-1(V), controls and indicators - dual satellite coverage configuration. - (Continued)

Control or indicator marking	Type	Function
PHONE	Telephone jack	Provides access to sound-powered intercom circuit with both antenna pedestals
ANTENNA NO 2, AZIMUTH ERROR	Indicator (red)	When lit, indicates a discrepancy between the AZIMUTH command angle and the actual antenna position. Lamp is extinguished after antenna moves to the commanded azimuth angle
ANTENNA NO 2, ELEVATION, ERROR	Indicator (red)	When lit, indicates a discrepancy between the ELEVATION command angle and the actual antenna position. Lamp is extinguished after antenna moves to the commanded elevation angle
MODE READY, AUTO	Indicator (amber)	When lit, indicates that Own Ship Heading inputs are present at the antenna control unit
MANUAL, AUTO	Toggle switch	In AUTO position, Own Ship Heading inputs are used as a reference to provide true azimuth command signals. In MANUAL position, the ship's 115 VAC, 400 Hz power is used to provide a reference for azimuth command signals that are only relative to the heading of the ship
READY, MANUAL	Indicator (amber)	When lit, indicates that the ship's 115 VAC, 400 Hz power is available at the antenna control unit
ANTENNA NO 1, AZIMUTH, ERROR	Indicator (red)	When lit, indicates a discrepancy between the AZIMUTH command angle and the actual antenna position. Lamp is extinguished after antenna moves to the commanded azimuth angle

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TABLE II. Antenna control C-9597( )/WSC-1(V), controls and indicators - dual satellite coverage configuration. - (Continued)

Control or indicator marking	Type	Function
ANTENNA NO 1, ELEVATION ERROR	Indicator (red)	When lit, indicates a discrepancy between the ELEVATION command angle and the actual antenna angle. Lamp is extinguished after antenna moves to the commanded elevation angle
2A	Fuse holder	Contains 2-ampere fuse that protects the 28 VDC input circuits of the antenna control unit
S4	Rotary switch (located inside antenna control unit and accessible only by removing cover)	In position 1, configures antenna control unit for single satellite coverage In position 2, configures antenna control unit for dual satellite coverage by making antenna No. 2 separately controlled.

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3.1.1.4.6 Telephone circuit. The sound-power telephone circuit from the pedestal shall terminate on the front panel of the antenna control unit.

3.1.1.4.7 Weight. The weight of the antenna control unit shall not exceed 8.2 kg (18 lb.).

3.1.1.5 Antenna group interfaces. Antenna group interfaces are as specified in 3.1.1.5.1 through 3.1.1.5.2.4.

3.1.1.5.1 Mechanical interfaces. Mechanical interfaces are as specified in 3.1.1.5.1.1 through 3.1.1.5.1.5.

3.1.1.5.1.1 Pedestal. The pedestal mechanical interfaces are as shown in the Antennas 1A1 and 1A2, Installation Drawing figure and the Typical Pedestal Cover Plate figure of NAVELEX 0967-LP-614-0010.

3.1.1.5.1.2 Amplifier-filter unit. The amplifier-filter mechanical interfaces are as shown in the Amplifier-Filter, Bandpass 1A3 and 1A4 Installation Drawing figure of NAVELEX 0967-LP-614-0010.

3.1.1.5.1.3 Switching unit. The switching unit mechanical interfaces are as shown in the Switching Unit 1A5, Installation Drawing figure of NAVELEX 0967-LP-614-0010.

3.1.1.5.1.4 Antenna control unit. The antenna control unit mechanical interfaces are as shown in FIGURE 1 and FIGURE 2, and in the Antenna Control 1A6, Installation Drawing figure of NAVELEX 0967-LP-614-0010 as modified by 3.1.1.4.1.

3.1.1.5.1.5 Dry air interface. The dry air mechanical interfaces are as shown in the Pressurized Dry-Air Function, Block Diagram figure of NAVELEX 0967-LP-614-0010.

3.1.1.5.2 RF and electrical interfaces. RF and electrical interfaces shall be as specified in 3.1.1.5.2.1 through 3.1.1.5.2.4.

3.1.1.5.2.1 Pedestal. The pedestal RF and electrical interfaces are as shown in the Antenna Pedestal Power and Control Circuits (1A1A2 and 1A2A2), Schematic Diagram figure and the Antenna 1A1 and 1A2, Installation Drawing figure of NAVELEX 0967-LP-614-0010.

3.1.1.5.2.1.1 Drive motor interface. The power for the azimuth axis motor and the elevation axis motor shall be 208 V, 3-phase, 4-wire (wye), 50 to 60 Hz.

3.1.1.5.2.1.2 Azimuth drive direction. When the antenna pedestal is installed on platforms with cabling in existence from a previously installed antenna pedestal, the sense of azimuth rotation shall be the same as that of the first system without changing any cabling.

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3.1.1.5.2.1.3 Elevation drive direction. When the antenna pedestal is installed on platforms with cabling in existence from a previously installed antenna pedestal, the sense of elevation rotation shall be the same as that of the first system without changing any cabling.

3.1.1.5.2.1.4 Synchro output interface. Each axis of the pedestal shall contain one TYPE 26V-08-CT or TYPE 26V-11-CT control transformer in accordance with MIL-S-20708.

3.1.1.5.2.1.5 Azimuth error direction. With electrical angle of zero degrees applied at pedestal connector J2 pins K, J, and H (S1, S2, and S3 respectively), a pedestal clockwise position error shall result in R1/R2 error voltage at pins F and G respectively. For this direction of error, the voltage of pin F shall be in phase with the voltage at pin K.

3.1.1.5.2.1.6 Elevation error direction. With electrical angle of zero degrees applied at pedestal connector J2 pins A, B, and C (S1, S2, and S3 respectively), a pedestal UP position error shall result in R1/R2 error voltage at pins E and D respectively. For this direction of error, the voltage at pin E shall be in phase with the voltage at pin A.

3.1.1.5.2.2 Amplifier-filter unit. The amplifier-filter unit RF and electrical interfaces are as shown in the Amplifier-Filter, Bandpass (1A3, 1A4), Schematic Diagram figure and the Amplifier-Filter, Bandpass 1A3 and 1A4, Installation Drawing figure of NAVELEX 0967-LP-614-0010.

3.1.1.5.2.3 Switching unit. The switching unit RF and electrical interfaces are as shown in the Switching Unit (1A5), Schematic Diagram figure and the Switching Unit 1A5, Installation Drawing figure of NAVELEX 0967-LP-614-0010 as modified by 3.1.1.3 and 3.1.1.4 for the satellite coverage configurations.

3.1.1.5.2.4 Antenna control unit. The antenna control unit electrical interfaces are as shown in the Antenna Control (1A6), Schematic Diagram figure and the Antenna Control 1A6, Installation Drawing figure of NAVELEX 0967-LP-614-0010 as modified by 3.1.1.4 for the satellite coverage configurations.

3.1.1.6 Interchangeability requirements. The replaceable assemblies specified in 3.1.1.6.1 through 3.1.1.6.4 provided under a single contract or order shall be fully interchangeable. The replaceable assemblies shall be interchangeable on a noninterference basis with all such assemblies provided on previous contracts or orders.

3.1.1.6.1 Antenna unit. Within the antenna unit, the replaceable assemblies shall be those specified in a through f:

a. Radiating assembly
b. Drive motor assembly
c. Slip ring assembly
d. Synchro assembly
e. RF rotary joint
f. Azimuth bearing

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3.1.1.6.2 Amplifier-filter unit. Within the amplifier-filter unit, the replaceable assemblies shall be those specified in a and b:

a. Diplexer
b. Preamplifier

3.1.1.6.3 Switching unit. Within the switching unit, the replaceable assemblies shall be those specified in a through c:

a. Transmit-transfer coaxial relay
b. All other coaxial relays
c. 28 VDC power supply

3.1.1.6.4 Antenna control unit. Within the antenna control unit, the replaceable assemblies shall be those specified in a and b:

a. Electronic switch printed circuit board
b. Scott-T printed circuit board

3.1.1.7 Antenna group intermodulation. All odd ordered intermodulation products in the receive pass band shall be at or below -115 dBm at the input to the preamplifier when two input signals, each at a power of 100 W, are combined and simultaneously applied to the input of the antenna group (switching unit). Both unmodulated and phase-shift-keyed modulated carriers at data rates of up to 32,000 bps shall be used with carrier frequencies in the transmit pass band for worst case combinations such that intermodulation products fall in the receive pass band.

3.2 First article. One antenna group from the first five production groups shall be subjected to first article inspection (see 4.3).

3.3 Quantitative reliability requirements. The upper test mean-time-between-failure (MTBF) of the antenna group (θ_0 as defined by MIL-STD-781) shall be 3000 hours and the lower test MTBF (θ_1 as defined by MIL-STD-781) shall be 1500 hours.

3.4 Maintainability. Ease of maintenance, trouble-shooting and repair shall be a primary design consideration. Panels and doors which must be opened for maintenance or inspection more often than monthly shall have a maximum time to open of five seconds. Maintenance and repair shall be accomplished to the maximum extent practicable when the system is operational. The antenna group shall have a mean corrective maintenance time (M_{ct}) of 2.5 hours and a maximum corrective maintenance time (Mmaxct) of 7.5 hours at a 95 percent confidence level when repair is accomplished by replacement of modules and chassis mounted components (this includes electronic and electro-mechanical parts, but excludes gears, bearings, and radiating assemblies). The M_{ct} includes localization, isolation, disassembly, interchange, reassembly, alignment and checkout of all maintenance tasks. This excludes all preventive maintenance associated with the antenna group.

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3.5 Environmental requirements. The above-deck and below-deck equipment shall conform to the requirements specified in 3.5.1 through 3.5.2.12.

3.5.1 Above-deck equipment. The above-deck equipment (antenna and amplifier-filter units) shall conform to the environmental requirements specified in 3.5.1.1 through 3.5.1.13.

3.5.1.1 Temperature range. The above-deck equipment shall operate from -28°Celsius (C) to +65°C and have a nonoperating (storage) range of -62°C to +71°C.

3.5.1.2 Salt fog. The above-deck equipment shall be resistant to the corrosive effects of salt sea atmosphere.

3.5.1.3 Altitude. The above-deck equipment shall survive nonoperating storage and transport in a low pressure environment equivalent to 15,000 feet (ft) above sea level. The above-deck equipment shall operate normally at pressure equivalent to 5000 ft above sea level.

3.5.1.4 Wind. The above-deck equipment shall operate normally in wind speeds up to 75 knots. The above-deck equipment, nonoperating, shall survive wind speeds of 100 knots.

3.5.1.5 Ice loading. The above-deck equipment shall operate normally with ice loading of up to 4.5 pounds per square foot.

3.5.1.6 Sea state. The above-deck equipment shall operate normally under sea state five as specified in DOD-STD-1399 Sections 301 and 302. The above-deck equipment, nonoperating, shall survive sea state eight.

3.5.1.7 Shock. The above-deck equipment shall conform to the shock requirements of MIL-S-901 for Grade A, Class 1, Type A.

3.5.1.8 Vibration. The above-deck equipment shall conform to the vibration requirement of MIL-STD-167-1, Type I. The maximum frequency shall be 50 Hz. The equipment shall comply with the vibration requirements without internal or external resilient mountings.

3.5.1.9 Electromagnetic compatibility. The above-deck equipment shall be self-compatible and meet the following requirements of MIL-STD-461 for class A4 equipment: CE01, CE03, RE01, RE02, RS01, RS02, RS03, CS01, CS02, CS03, CS04, CS05, and CS06 (at 250 V spike).

3.5.1.10 Enclosure. The above-deck equipment shall be spraytight in accordance with MIL-STD-108.

3.5.1.11 Humidity. For above-deck equipment, the humidity requirements shall be in accordance with MIL-STD-810, Test Method 507.2, Procedure 1, Cycle 1, for non-hazardous equipment. The equipment shall withstand humidity environments up to 95 percent maximum.

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3.5.1.12 Inclination. The above-deck equipment shall conform to the Inclination paragraph of MIL-E-16400 for surface ships.

3.5.1.13 Electromagnetic pulse (EMP) hardening. The above-deck equipment shall be hardened or shall be intrinsically hard against the EMP resulting from on exoatmospheric nuclear burst. The equipment shall not exhibit any permanent malfunction or degradation of any requirements of the equipment as specified herein, after being subjected to the RS05 test signal requirement of MIL-STD-461 for Class A4 equipment.

3.5.1.14 Nuclear effects. The above-deck equipment shall conform to the requirements of 3.5.2.14.1 through 3.5.1.14.3.

3.5.1.14.1 Transient radiation effects on electronics (TREE). The above-deck equipment shall operate under a TREE environment and survive the nuclear hardness levels specified in Enclosure (1) to Ref (b) of NAVMATINST 3401.1 (18 April 1985) without permanent damage, or interruption of service beyond the duration of the transient. Degradation due to neutrons or total dose shall not cause the equipment to perform below allowable limits for operation as specified herein.

3.5.1.14.2 Nuclear air blast. The above-deck equipment shall comply with the nuclear air blast requirements as specified in Enclosure (1) to Reference (b) of NAVMATINST 3401.1 (18 April 1985) without permanent damage, or interruption of service beyond the duration of the air blast.

3.5.1.14.3 Nuclear thermal radiation exposure. The above-deck equipment shall comply with the nuclear thermal radiation requirements as specified in Enclosure (1) to Reference (b) of NAVMATINST 3401.1 (18 April 1985) without permanent damage, or interruption of service beyond the duration of the thermal radiation.

3.5.2 Below-deck equipment. The below-deck equipment (switching and antenna control units) shall conform to the environmental requirements specified in 3.5.2.1 through 3.5.2.11.

3.5.2.1 Temperature range. The below-deck equipment shall operate from 0°C to +50°C and have a nonoperating (storage) range of -62°C to +71°C.

3.5.2.2 Salt fog. The below-deck equipment shall be resistant to the corrosive effects of salt sea atmosphere.

3.5.2.3 Altitude. The below-deck equipment shall survive nonoperating storage and transport in a low pressure environment equivalent to 15,000 ft above sea level. The below-deck equipments shall operate normally at pressure equivalent to 5000 ft above sea level.

3.5.2.4 Sea state. The below-deck equipment shall operate normally under sea state five as specified in DOD-STD-1399, Sections 301 and 302. The below-deck equipment shall survive, nonoperating, sea state eight.

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3.5.2.5 Shock. The below-deck equipment shall conform to the shock requirements of MIL-S-901 for Grade A, Class 1, Type A.

3.5.2.6 Vibration. The below-deck equipment shall conform to the vibration requirement of MIL-STD-167-1, Type I. The maximum frequency shall be 50 Hz. The equipment shall comply with the vibration requirements without internal or external resilient mountings.

3.5.2.7 Electromagnetic compatibility. The below-deck equipment shall be self-compatible and meet the following requirements of MIL-STD-461 for class A4 equipment: CE01, CE03, RE01, RE02, RS01, RS02, RS03, CS01, CS02, CS03, CS04, CS05, CS06 (at 250 volts spike).

3.5.2.8 Enclosure. The below-deck equipment shall be drip proof to 45 degrees in accordance with MIL-STD-108.

3.5.2.9 Humidity. For below-deck equipment the humidity requirements shall be in accordance with MIL-STD-810, Test Method 507.2 procedure 1, for non-hazardous equipment. The equipment shall withstand humidity environments up to 95 percent maximum.

3.5.2.10 Inclination. The below-deck equipment shall conform to the Inclination paragraph of MIL-E-16400 for surface ships.

3.5.2.11 EMP hardening. The below-deck equipment shall be hardened or shall be intrinsically hard against the EMP resulting from on exoatmospheric nuclear burst. The equipment shall not exhibit any permanent malfunction, degradation of performance, or deviation from any requirement of the equipment specified herein after being subjected to the CS11 test signal requirement of MIL-STD-461 for Class A4 equipment.

3.5.2.12 Nuclear TREE hardening. The below-deck equipment shall operate under a TREE environment and survive the nuclear hardness levels specified in Enclosure (1) to Reference (b) of NAVMATINST 3401.1 without permanent damage or interruption of service beyond the duration of the transient. Degradation due to neutrons or total dose shall not cause the equipment to perform below allowable limits for operation specified herein.

3.6 Parts, materials and processes. The antenna group shall conform to the requirements specified in 3.6.1 through 3.6.6.

3.6.1 Selection of parts. Selection of parts, materials, processes and standards shall be in accordance with MIL-E-16400 and as stated herein. Parts shall be derated in usage to provide a minimum safety factor as specified in Appendix A. Parts shall be rated at 125°C minimum, or higher, if required by circuit application. Plastic encapsulated micro-circuits shall not be used.

3.6.2 Parts control. The parts to be incorporated in the antenna group shall be controlled in accordance with MIL-STD-965, Procedure I.

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3.6.3 Parts rescreening. All semiconductor, microelectronic, and hybrid devices shall be rescreened at -55°C, +25°C, and +125°C. The applicable device specification parameters for each device shall be measured at each temperature.

3.6.4 Metal parts. Metal parts used in fabrication of this equipment shall be of the corrosion-resistant type described in MIL-E-16400, or shall be plated to provide resistance to corrosion.

3.6.5 Fungus. The equipment shall be resistant to fungi and shall not be adversely affected by fungi under conditions favorable for fungus development (for example, high humidity, warm atmosphere and presence of inorganic salt) in accordance with the Fungus paragraph of MIL-E-16400.

3.6.6 Anti-seize tape. Anti-seize tape as specified in MIL-T-27730 shall be used for all pipe-threaded hardware into aluminum housings.

3.7 Design and construction. The antenna group shall conform to the requirements specified in 3.7.1 through 3.7.22.

3.7.1 Accessibility. Exterior doors, hinged covers, and slide drawers shall have automatic latch and quick release devices which must be actuated before they can be opened, and stops to lock them in the open position for each maintenance configuration. The design shall be such that sufficient clearance is provided in each maintenance configuration to allow required tasks to be performed without undue discomfort, danger, or effort. Access doors, if required, shall be nonstructural and installed with the minimum number of captivated fasteners. Easy access to all components for maintenance shall be provided to the maximum extent practicable. The controls shall be readily accessible and simple to operate. The antenna group shall be maintainable by a two-person crew over any 12 hour period.

3.7.2 Maintenance. The maintenance concept for repair of the antenna group shall be organizational replacement of plug-in type modules, printed wiring boards (PWB), drive motors, limit switches. Components and piece parts incident to, but not part of, modules or PC boards may be replaced by a technician. Circuits shall be modularized so that similar functions shall use interchangeable modules. Replacement modules shall not require adjustment when integrated into the configuration. Connectors shall be located and spaced to allow ease of handling by maintenance personnel wearing gloves. Input and output connectors for the antenna control unit shall be provided at the rear of the chassis; however, maintenance shall be accomplished only from the front of the cabinet assembly when the antenna control unit drawer is withdrawn.

3.7.3 Safety criteria. The antenna group design shall conform to the safety criteria requirements specified in 3.7.3.1 through 3.7.3.5.

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3.7.3.1 Critical controls. Critical controls, the activation of which may cause damage to the antenna group, injury to personnel, or degradation of system function, shall be designed and located so that the controls are not susceptible to being moved accidentally.

3.7.3.2 Acoustic noise. Sound levels produced by the antenna group shall not exceed an A-weighted sound pressure of 80 dB when measured 3 ft from the equipment.

3.7.3.3 Weight marking. Antenna group components weighing in excess of 35 pounds shall have the weight clearly marked on the external surface of the antenna group and readily visible during installation and removal. The marking shall be as permanent as the normal life expectancy of the unit to which it is applied.

3.7.3.4 Leakage current. The equipment leakage current shall not exceed 5 milliamperes (mA) at 115 V, 400 Hz, single phase or 440 V, 50 to 60 Hz, three phase alternating current (AC) power.

3.7.3.5 115 VAC equipment power connections. 115 VAC power cable connections within the equipment shall have the conductor and color assignments shown in TABLE III. When connectors are used, the pin designation shall be as shown in TABLE III. Black paired with white or brown paired with light blue conductor insulation color shall be maintained from the equipment input connections to all components having the same voltage and frequency as the input power.

TABLE III. 115 VAC equipment power connections.

Pin designation	Conductor assignment	Insulation color code	
		National	International
A	115 VAC return	White	Light blue
B	Safety ground	Green	Green with yellow strip
C	115 VAC high (supply)	Black	Brown

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3.7.4 PWB protection and PWB fabrication. PWB fabrication and protection shall be as specified in 3.7.4.1 and 3.7.4.2 below.

3.7.4.1 PWB protection. With power applied to the antenna group, removal or insertion of PWBs shall not damage the assembly or any other part of the antenna group. If required, to prevent damage to an assembly, the equipment shall be keyed to prevent incorrect insertion.

3.7.4.2 PWBs. PWBs shall be fabricated in accordance with the requirements of MIL-STD-275 and MIL-P-55110. PWBs and chassis shall be marked to key and facilitate insertion of the PWB into the correct position. No damage shall occur if a PWB is inserted into an incorrect position.

3.7.5 Power supply protection. Fault conditions ranging from open circuits to short circuits shall cause no damage to the power supply.

3.7.6 Installation kits and mating connectors. A complete set of mating halves of connectors and covers or protective caps shall be provided. The mating connectors shall comply with the Improper application prevention paragraph of MIL-E-16400. Any special equipment required for installation shall be provided.

3.7.7 Electrostatic discharge protection requirements. Components and modules with inherent susceptibility to the effects of electrostatic discharge shall be properly protected in conformance to the requirements of DoD-STD-1686 and DoD-HDBK-263.

3.7.8 Material flammability. Material used as part of the equipment shall conform to MIL-STD-454, Requirement 3.

3.7.9 Overcurrent protection. Electrical overcurrent protection shall conform to the Circuit breakers, fuses and fuseholders, and Electrical overload protection paragraphs of MIL-E-16400. When fuses are used, front panel blown fuse indicators shall be provided. When circuit breakers are used, a visible status indicator shall be provided.

3.7.9.1 Overcurrent device location. A fuse shall be connected in series with each ungrounded power conductor as determined by the NFPA 70-1984 for grounded source power and shall be located on the load side of the power switch.

3.7.10 Thermal design. Equipment thermal design shall be in accordance with the paragraphs of MIL-E-16400 specified in a through e:

a. Thermal design
b. Heat removal
c. Forced air cooling
d. Ventilation openings
e. Cooling during maintenance

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If equipment forced air cooling is required, all air intakes shall be covered by a suitable air filter which can be cleaned with soap and water.

3.7.10.1 High temperature protection. The equipment shall employ fail-safe devices which protect the equipment from damage in the event of loss of cooling air or in the event of excessively high ambient temperatures. The fail-safe devices shall provide protection while the equipment is in the standby mode. An override capability shall be provided.

3.7.11 Handles. Handles shall be provided on the front panel of the equipment to ensure safe and easy removal of the equipment from the mounting. The handles shall have no sharp edges. Handles shall not protrude more than 5.08 cm (2 in.) from the front panel surface. All controls on the front panel shall not protrude beyond the handles.

3.7.12 Locking devices. Locking devices or materials required on internal adjustment controls, to prevent the need for readjustment of the controls after sustained periods of shock and vibration, shall conform to MIL-STD-454, Requirements 12 and 28.

3.7.13 Mounting holes. All tapped aluminum mounting holes (except grounding points and pipe threads) shall contain inserts.

3.7.14 Finish. The exterior of the antenna and amplifier-filter units shall be painted gray in accordance with MIL-P-24441. The exterior of the antenna control and switching units shall be painted in accordance with MIL-E-15090, Class 2 (semi gloss), Type III (baking) enamel. Painting of the front panel and edges of the equipment shall be in accordance with the Painting paragraph of MIL-E-16400. Epoxy paint conforming to MIL-C-22751 may be used in lieu of enamel. The remaining surfaces of the equipment shall be treated with chemical film in accordance with MIL-C-5541.

3.7.15 Electrical cable protection. The equipment design shall preclude damage to electrical cabling during all conditions of assembly, removal, insertion, and tilting of the equipment.

3.7.16 RF connectors. RF connectors shall exclude connectors using ferromagnetic materials and shall be provided in accordance with and conform to the requirements of the Connectors, radio frequency paragraph of MIL-E-16400 and MIL-C-39012.

3.7.17 Controls, indicators, and panel layouts. The design of controls and indicators and their arrangement on operator and maintenance panels shall be in accordance with the Controls, indicators, and panel layout paragraph of MIL-E-16400. The abbreviations of MIL-STD-12 shall be used for equipment labeling where applicable.

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3.7.18 Unacceptable materials. The equipment design shall comply with the Unacceptable materials paragraph of MIL-E-16400. The equipment design shall not include the materials specified in a through f:

a. Flammable or combustible materials
b. Polychlorinated biphenyls
c. Asbestos and asbestos compounds
d. Fragile or brittle materials
e. Beryllium and beryllium compounds unless so identified
f. Lithium and lithium compounds unless specifically approved by the procuring activity

3.7.19 Derating criteria. The derating criteria of NAVSEA 0967-LP-597-1011 shall be utilized in the design of the equipment.

3.7.20 Power supply design. Guidance for the design and construction of power supplies is provided in NAVMAT P4855-1 and MIL-HDBK-241. Power density in excess of 2 W per cubic inch shall require the approval of the Government. The junction temperature of semiconductor devices shall not exceed +110°C under worst case conditions. The power supply shall have a reserve capacity of at least 40 percent.

3.7.21 Soldering. Soldering shall conform to the requirements of WS-6536E with SCN 1 and SCN 2.

3.7.22 Magnetic materials. Non-magnetic materials shall be used for all parts except where magnetic materials are essential, in which case a listing of the size, quantity, and application of magnetic materials with a permeability greater than 2.0 shall be provided to the Government for approval prior to engineering release.

3.8 Identification and marking. Antenna group identification and marking shall conform to the requirements specified in 3.8.1 through 3.8.4.

3.8.1 Antenna group identification. Marking of all antenna groups shall be in accordance with the requirements of MIL-P-15024.

3.8.2 Visibility. When assembled, identification markings of parts shall be readily visible. Auxiliary markings, such as on access panels, shall be provided so as to identify those parts where labeling is not readily visible.

3.8.3 Readability of identification plates. Nomenclature plates and serial numbers shall be located so they are readable after antenna group installation. They shall be located on the front surface of the control panel of below-deck equipment and on the side of above-deck equipment.

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3.8.4 Identification of wiring. Wiring shall be identified by color-coding, or by other means, such as marking of terminal at both ends of each lead. Color-coding for chassis wiring shall conform to the requirements of MIL-STD-681.

3.9 Antenna group power. The antenna group shall conform to power requirements in accordance with DOD-STD-1399, Section 300 and as specified in 3.9.1 through 3.9.2.

3.9.1 Type I power. 440 V, 3 phase, 3 wire delta, 50 Hz to 60 Hz, 2.5 Kilovoltampere (KVA) (nominal), 0.92 power factor.

3.9.2 Type II power. 115 V, 1 phase, 2 wire, 400 Hz, 0.3 VA (nominal), 1.0 power factor.

3.10 Own ship heading. The antenna group shall utilize Own Ship Heading inputs for control of azimuth pointing angles. The inputs shall be 115 V, 400 Hz, 360 degrees per revolution synchro signals in accordance with DOD-STD-1399, Section 702. All interfaces with the own ship data set shall be in accordance with DOD-STD-1399, Section 702 and the Synchros paragraph of MIL-E-16400.

3.11 Workmanship. Workmanship shall be as specified in 3.11.1 and 3.11.2.

3.11.1 General workmanship. Workmanship shall conform to the Workmanship paragraph of MIL-E-16400.

3.11.2 Workmanship screen. All equipment produced in accordance with this specification shall withstand a random vibration screen and a temperature screen.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall conform to the requirements of Section 3 and Section 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of

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assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government verification. All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of operations to determine that practices, methods, and procedures of the written quality program are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to ensure compliance with all inspection requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or product conformance to prescribed criteria has been demonstrated.

4.1.3 Quality assurance terms and definitions. Quality assurance terms and definitions used in this specification shall be as defined in MIL-STD-109.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

1. Production inspection (Group A) (see 4.4.1)
2. Production control inspection (Group B) (see 4.4.2)
3. Environmental inspection (Group C) (see 4.4.3)

c. Inspection of preparation for delivery (see 4.9).

4.3 First article inspection. Unless otherwise specified (see 6.2), one out of the first five production antenna groups shall be subjected to first article inspection. First article inspection shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in TABLE IV.

4.4 Quality conformance inspection. Quality conformance inspections shall be as specified in 4.4.1 through 4.4.3.

4.4.1 Production inspection (Group A). Production inspection shall be made on every antenna group offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process, such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. The antenna groups shall be subjected, as a minimum, to all production tests specified in TABLE IV, Group A, including subparagraphs.

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TABLE IV. Examinations and tests.

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Surface examination						
Weight	3.1.1.1.3, 3.1.1.2.4, 3.1.1.3.12, 3.1.1.4.7	4.5.1	X	X		
Size and interfaces	3.1.1.4.2, 3.1.1.5.1.1 through 3.1.1.5.1.5	4.5.1	X	X		
Parts and materials	3.1.1.1.1.9, 3.1.1.1.2.5, 3.1.1.1.2.10, 3.1.1.1.2.12 through 3.1.1.1.2.14, 3.1.1.1.6, 3.1.1.1.10, 3.1.1.1.13, 3.1.1.2.1.10, 3.1.1.2.2.10, 3.1.1.3.6, 3.6,3.7.8, 3.7.18 3.7.22	4.5.1	X	X		
Design and construction	3.7.4 through 3.7.7, 3.7.9 through 3.7.13, 3.7.15 through 3.7.17, 3.7.19 and 3.7.20	4.5.1	X	X		
Finish	3.1.1.1.1.10, 3.1.1.1.1.11, 3.1.1.1.2.12, 3.1.1.2.1.9, 3.1.1.2.2.9, 3.7.14	4.5.1	X	X		
Marking	3.1.1.1.1.13, 3.1.1.1.14, 3.1.1.1.5, 3.8.1 through 3.8.4	4.5.1	X	X		

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TABLE IV. Examinations and tests. - (Continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Bonding and grounding	3.1.1.2.3	4.5.1	X	X		
Safety	3.1.1.1.12, 3.7.3 through 3.7.3.5	4.5.1	X	X		
Workmanship	3.7.21, 3.11.1	4.5.1	X	X		
Operating test		4.5.2	X	X		
Pre- and post-test functional verification		4.5.3	X			X
Shock	3.5.1.7, 3.5.2.5	4.5.14	X			X
Vibration	3.5.1.8, 3.5.2.6	4.5.15	X			X
Inclination	3.5.1.12, 3.5.2.10	4.5.17	X			X
Temperature	3.5.1.1, 3.5.2.1	4.5.4	X			X
Humidity	3.5.1.11, 3.5.2.9	4.5.5	X			X
Enclosure	3.5.1.10, 3.5.2.8	4.5.16	X		X	X
Altitude	3.5.1.3, 3.5.2.3	4.5.11	X			X
Salt fog	3.5.1.2, 3.5.2.2	4.5.6	X			X
Wind	3.5.1.4	4.5.7	X			X
Ice loading	3.5.1.5	4.5.8	X			X
Sea state	3.5.1.6, 3.5.2.4	4.5.13	X			
EMP	3.5.1.13, 3.5.2.11	4.5.9.1	X			
Electromagnetic compatibility	3.5.1.9, 3.5.2.7	4.5.9	X			
Workmanship screen	3.11.2	4.6	X	X		
Leakage current	3.7.3.4	4.5.12	X		X	
Reliability	3.3	4.7	X		X	
Nuclear effects	3.5.1.14, 3.5.2.12	4.5.10	X			

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TABLE IV. Examinations and tests. - (Continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Maintainability	3.4, 3.7.1 and 3.7.2 3.1.1.3.7.1 through 3.1.1.3.7.2	4.8	X			
Interchangeability	3.1.1.6.1 through 3.1.1.6.4	4.9	X			
Prime power	3.1.1.3.7.1 through 3.1.1.3.7.2	4.5.1	X		X	
	3.9 through 3.9.2	4.5.1	X		X	
RF power	3.1.1.1.1.1, 3.1.1.1.2.4, 3.1.1.1.7, 3.1.1.2.2.7, 3.1.1.3.1.3, 3.1.1.3.2.4, 3.1.1.3.3.5, 3.1.1.3.4.5	4.5.1	X		X	
Frequency range	3.1.1.1.1.2	4.5.1	X		X	
Intermodulation	3.1.1.1.1.8, 3.1.1.1.2.15, 3.1.1.2.1.5, 3.1.1.2.2.8, 3.1.1.3.9, 3.1.1.7	4.5.1	X		X	
VSWR	3.1.1.1.1.6, 3.1.1.1.2.4, 3.1.1.1.8, 3.1.1.2.1.6, 3.1.1.2.1.7, 3.1.1.2.2.6, 3.1.1.3.1.4, 3.1.1.3.2.3, 3.1.1.3.3.4, 3.1.1.3.4.4	4.5.1	X		X	

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TABLE IV. Examinations and tests. - (Continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Isolation	3.1.1.2.2.5, 3.1.1.3.1.6, 3.1.1.3.3.1, 3.1.1.3.4.1,	4.5.1	X		X	
Insertion loss	3.1.1.1.2.4, 3.1.1.2.2.1, 3.1.1.2.2.2, 3.1.1.2.2.3, 3.1.1.2.2.4, 3.1.1.3.1.5, 3.1.1.3.2.2, 3.1.1.3.3.3, 3.1.1.3.4.3	4.5.1	X		X	
Noise figure	3.1.1.2.1.4	4.5.1	X		X	
RF and electrical interfaces	3.1.1.5.2.1, 3.1.1.5.2.1.1, 3.1.1.5.2.1.4, 3.1.1.5.2.2 through 3.1.1.5.2.4	4.5.1	X		X	
Auxiliary receive output	3.1.1.3.8	4.5.1	X		X	
Antenna pattern	3.1.1.1.1.3 through 3.1.1.1.1.5, 3.1.1.1.1.7	4.5.1	X		X	
Wow	3.1.1.1.2.4	4.5.1	X		X	
Preamplifier gain	3.1.1.2.1.2, 3.1.1.2.1.3	4.5.1	X		X	
Keyline controls	3.1.1.3.5 through 3.1.1.3.5.2	4.5.1	X		X	
Burn out protection	3.1.1.2.1.8	4.5.1	X		X	
Antenna selection	3.1.1.4.1	4.5.1	X		X	
Antenna override	3.1.1.4.4	4.5.1	X		X	
Controls and indicators	3.1.1.4.3, 3.1.1.3.13	4.5.1	X		X	
Antenna control configuration	3.1.1.4.5 through 3.1.1.4.5.2, 3.1.1.4.6	4.5.1	X		X	

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TABLE IV. Examinations and tests. - (Continued)

Examination or test	Requirement paragraph	Test paragraph	First article inspection	Quality conformance inspection		
				Group A	Group B	Group C
Angular coverage	3.1.1.1.2.3	4.5.1	X		X	
Angular velocity	3.1.1.1.2.6	4.5.1	X		X	
Angular accuracy	3.1.1.1.2.2, 3.1.1.1.2.7, 3.1.1.1.2.8, 3.1.1.1.2.9	4.5.1	X	X		
Direction of rotation	3.1.1.5.2.1.2, 3.1.1.5.2.1.3, 3.1.1.5.2.1.5, 3.1.1.5.2.1.6	4.5.1	X	X		
Sector switching	3.1.1.1.2.11	4.5.1	X		X	
Switching time	3.1.1.3.1.2, 3.1.1.3.2.1, 3.1.1.3.3.2, 3.1.1.3.4.2	4.5.1	X		X	
Stops	3.1.1.1.11	4.5.1	X	X		
Swept volume	3.1.1.1.4	4.5.1	X		X	
Air pressure system	3.1.1.3.10 through 3.1.1.3.11	4.5.1	X		X	

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4.4.2 Production control inspection (Group B). Production control inspection shall be conducted on a sampling basis, where sample size is determined using special inspection level S-3 as specified in MIL-STD-105 for normal, tightened, and reduced inspection. Failure of any unit within the sample to conform to prescribed requirements shall constitute a failure and shall be cause for rejection of the inspection lot. Production control inspection shall include those examinations and tests specified in TABLE IV, Group B and shall be performed on receiving sets that have passed production inspection. The receiving sets shall satisfactorily conform to the requirements of production control inspection prior to release for shipment.

4.4.2.1 Rejected lots. If an inspection lot is rejected, the contractor may withdraw the lot from further inspection. The contractor may also rework a rejected lot to correct the defective units and reinspect the lot, using tightened inspection. Rejected lots shall be kept separate from new lots and shall not lose their identity.

4.4.3 Environmental inspection (Group C). Environmental inspection shall be performed on antenna groups which have been subjected to and have passed production inspection. Environmental testing shall consist, as a minimum, of the tests of TABLE IV, Group C.

4.4.3.1 Environmental inspection samples. Samples for the environmental inspection shall be selected without regard to their quality in accordance with a and b:

a. One sample from the first 25 production antenna groups fabricated (the first article test may be used to satisfy this requirement)
b. One sample from each successive 100 antenna groups, or fraction thereof, produced

4.4.4 Nonconformance corrective action and retest. If a sample antenna group fails the inspection specified in 4.4.2 or 4.4.3, the contractor shall immediately investigate the cause of failure and shall implement the corrective action necessary to correct the production antenna groups which were manufactured with the same conditions, materials, processes. The equipment shall then be retested.

4.4.5 Reinspection of production control and environmental test antenna groups. Unless otherwise specified in the contract (see 6.2), sample antenna groups which have been subjected to and have passed production control and environmental inspections may be accepted on the contract provided they are resubjected to, and pass, production inspection specified in 4.4.1 after repair of all visible damage.

4.5 Test methods. Test methods shall be as specified in 4.5.1 through 4.5.19.

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4.5.1 Test and examinations. Tests and examinations specified herein shall be conducted in accordance with the applicable inspection procedures of MIL-E-16400 and Government approved test procedures. It shall be the responsibility of the contractor to provide all cables, connectors, test instrumentation, test facilities, and other devices required to set up and perform all tests at the contractor's plant, or other place of test, to verify compliance with the requirements specified herein. When the antenna group is tested as a system, the contractor shall provide interconnecting cables adequate to simulate realistic installation conditions; that is, length, attenuation, and shielding shall be as identical as possible to actual installations.

4.5.2 Operating test. The antenna group shall be energized and subjected to an operating test to ensure the proper functioning and performance at an adequate number of points in the required range to demonstrate conformance to the requirements specified herein.

4.5.3 Pre- and post-test functional verification. Prior to proceeding with any of the Group C test methods, the test item shall be operated under standard ambient conditions, and a record shall be made of all data necessary to determine compliance with required performance. These data shall provide the criteria for checking satisfactory performance of the test item either during or at the conclusion of the test, or both, as required. When operation of the test item is required during the test exposure, the operation and performance checks shall be of sufficient duration or shall be repeated at appropriate intervals, to ensure a record of comprehensive comparative data for comparison with data recorded under standard ambient conditions. The test item shall be visually examined, and a record shall be made of any damage or deterioration resulting from the test. If a test chamber is used for the test, a visual examination of the test item within the chamber shall be performed at test conditions, when possible. Upon completion of the test, a visual examination of the test item shall be performed again after the test item has been returned to standard ambient conditions. Deterioration, corrosion, or change in tolerance limits of any internal or external parts which could in any manner prevent the test item from conforming to operational service or maintenance requirements, shall provide reason to consider the test item as having failed to withstand the conditions of the test. Leakage or degradation of impregnating compounds shall be considered damage and cause for rejection.

4.5.4 Temperature tests. The equipment shall be tested in accordance with MIL-STD-810, Test Method 501.2, Procedures I and II and Test Method 502.2, Procedures I and II to demonstrate conformance with 3.5.1.1 and 3.5.2.1.

4.5.5 Humidity test. The equipment shall be tested in accordance with MIL-STD-810, Test Method 507.2 to demonstrate conformance with 3.5.1.11 and 3.5.2.9.

4.5.6 Salt fog test. The equipment shall be tested in accordance with MIL-STD-810, Test Method 509.2 to demonstrate conformance with 3.5.1.2 and 3.5.2.2.

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4.5.7 Wind speed. The above-deck equipment shall be tested to demonstrate conformance with 3.5.1.4.

4.5.8 Ice loading. The above-deck equipment shall be tested in accordance with MIL-STD-810, Test Method 521.0 to demonstrate conformance with 3.5.1.5.

4.5.9 Electromagnetic environment effects (E^3) (including EMP). The equipment shall be tested in accordance with MIL-STD-462 to demonstrate conformance with 3.5.1.9 and 3.5.2.7.

4.5.9.1 EMP. The equipment shall be tested in accordance with MIL-STD-462 to demonstrate conformance with 3.5.1.13 and 3.5.2.11.

4.5.10 Nuclear effects testing. Verification of conformance of hardness to nuclear effects shall be demonstrated either by analysis or nuclear simulation tests to show compliance with 3.5.1.14 and 3.5.2.12. Electronic piece-parts and circuits for which TREE test data are available on the specific device can be verified by analysis. Test Plans submitted to the Government shall include the following: facility to be used; parts, circuits, or equipments to be tested; levels of test; diagnostic techniques; and description of data reduction methodology. For direct tests of shock, blast, or thermal effects, test plans shall include the test waveforms, interface circuits to be tested, test equipment, test fixtures, diagnostic techniques, and description of data reduction.

4.5.11 Altitude. The equipment shall be subjected to low pressure tests in accordance with MIL-STD-810, Test Method 500.2, Procedures I and II to demonstrate conformance with 3.5.1.3 and 3.5.2.3.

4.5.12 Leakage current. Leakage current shall be measured at 115 V, 400 Hz, single phase and 440 V, 50 Hz to 60 Hz, three phase.

WARNING

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. DO NOT TOUCH EXPOSED METAL SURFACES WITHOUT ADEQUATE ELECTRICAL SHOCK PROTECTION.

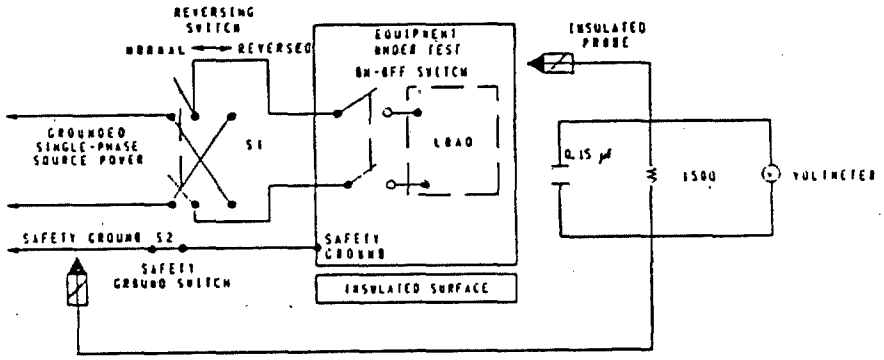
THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR DURING OR AS A RESULT OF THIS TEST.

4.5.12.1 Equipment test connections (115 VAC, 400 Hz, single phase). After power removal, each equipment directly connected to an external power source and units deriving power from the equipment shall be placed on an insulated surface. All safety ground conductors between the equipments and units deriving power from the equipment shall be intact. The safety ground conductor between the equipment and the source power shall be opened during the test. OBSERVE THE WARNING STATEMENT. The equipment shall be connected as shown in FIGURE 4 for a single-phase source.

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. S1 SW NORMAL. S2 SW CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING S2 SW OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT, RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. S1 SW REVERSED. ON-OFF SW ON. REPEAT STEP 4.
7. ON-OFF SW OFF. REPEAT STEP 4.
8. S2 SW CLOSED. S1 SW NORMAL.
9. REPEAT STEP 3 THRU 8 FOR EACH MODE OF OPERATION.
10. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT.



WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

FIGURE 4. Single phase test diagram for leakage current measurement.

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4.5.12.2 Measurement (115 VAC, 400 Hz, single phase). Leakage current shall be measured on the equipment in its normal operating configuration. Equipment controls in each operating mode shall be such that maximum power will be utilized during leakage current measurements. The leakage current shall be determined by the voltage-drop method. A true rms voltmeter shall be used. The voltage measured across the 1500-ohm resistor, when equal to 7.5 V, represents 5 mA of leakage current. The overall measurement error shall not exceed 5 percent. The probe shall be used on all external conducting parts such as case, connector housings, recessed calibration or adjustment controls, and control shafts with knobs removed. The voltage shall be measured from each part to ground for every combination of switch positions available in FIGURE 4. The open safety ground conductor shall be reconnected immediately after the test is completed.

4.5.12.3 Equipment test connections (440 VAC, 50 Hz to 60 Hz, three phase). Follow the same directions as paragraph 4.5.12.1 except use FIGURE 5 for a three-phase source.

4.5.12.4 Measurement (440 VAC, 50 Hz to 60 Hz, three phase). Follow the same directions as paragraph 4.5.12.2 except use FIGURE 5 for a three-phase source.

4.5.13 Sea state. An analysis shall be conducted to demonstrate conformance with 3.5.1.6 and 3.5.2.4.

4.5.14 Shock test. The equipment shall be tested in accordance with MIL-S-901, Grade A, Class 1, Type A to demonstrate conformance with 3.5.1.7 and 3.5.2.5. In addition to failure criteria specified in MIL-S-901, there shall be no more than 6 material fractures (tears or broken welds) of a length not to exceed 2 inches for each fracture of the antenna assembly and pedestal assembly. For the amplifier-filter unit, switching unit, and antenna control unit, any weld that is broken by shock testing shall be cause for rejection. Inoperability of the equipment during and after each blow of the shock test machine is allowable only if the system is self-correcting, requiring no operator interface. Malfunction requiring operator actions (that is, physical reset of circuit breakers, switches, and so forth) shall be cause for rejection.

4.5.15 Vibration test. The equipment shall be tested in accordance with MIL-STD-167-1, Type I without internal/or external resilient mountings to demonstrate conformance with 3.5.1.8 and 3.5.2.6.

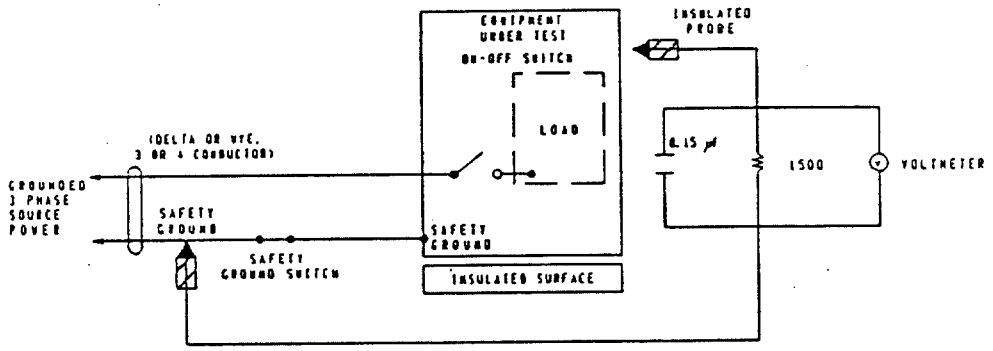
4.5.16 Enclosure test. The equipment shall be tested in accordance with MIL-STD-108 to demonstrate conformance with 3.5.1.10 and 3.5.2.8.

4.5.17 Inclination test. The equipment shall be tested in accordance with the Inclination test paragraph of MIL-E-16400 to demonstrate conformance with 3.5.1.12 and 3.5.2.10.

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GENERAL ORDER OF TEST:

1. SOURCE POWER OFF. CONNECT EQUIPMENT PER DIAGRAM.
2. ON-OFF SW OFF. SAFETY GROUND SWITCH CLOSED. CONNECT SOURCE POWER.
3. OBSERVE WARNING SAFETY GROUND SWITCH OPEN. ON-OFF SW ON.
4. FOR EACH PROBE POINT. RECORD VOLTMETER READING (CASE, CONNECTORS, CONTROLS, SHAFTS).
5. ON-OFF SW OFF. REPEAT STEP 4.
6. SAFETY GROUND SWITCH CLOSED.
7. REPEAT STEP 3 THRU 6 FOR EACH MODE OF OPERATION.
8. REMOVE SOURCE POWER. DISCONNECT EQUIPMENT.



WARNING

DO NOT TOUCH EXPOSED METAL SURFACES

THIS TEST MAY BE HAZARDOUS DUE TO THE UNGROUNDED CONDITION OF THE EQUIPMENT DURING THE TEST. THE UNITED STATES GOVERNMENT NEITHER ASSUMES NOR ACCEPTS RESPONSIBILITY FOR ANY INJURY OR DAMAGE THAT MAY OCCUR FROM THE USE OF THIS DIAGRAM FOR LEAKAGE CURRENT MEASUREMENT.

NOTES

1. ALL THREE PHASES SHALL BE CONNECTED DURING MEASUREMENT.
2. THE SAFETY GROUND CONDUCTOR SHALL NOT CARRY LOAD CURRENT.

FIGURE 5. Three phase test diagram for leakage current measurement.

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4.5.18 Fungus. Analysis of materials and parts shall be performed to demonstrate conpliance with 3.6.4.

4.5.19 Acoustic noise. The equipment shall be tested in accordance with MIL-STD-740 to demonstrate conformance with 3.7.3.2.

4.6 Workmanship screen. The antenna group shall be subjected to a random vibration screen and temperature screen as specified in 4.6.1 and 4.6.2 to demonstrate conformance with 3.11.2.

4.6.1 Random vibration screen. Prior to conducting temperature cycling, vibration shall be performed on all antenna groups. The antenna groups shall be mounted in their normal operational position (without shock isolators) and the vibration stress shall be applied vertical to this position for an accumulated time of 10 minutes. The vibration shall be in accordance with FIGURE 6. The control accelerometer shall be located next to one of the mounting points of the item under test. The acceleration spectral density (ASD) shall be controlled by an accelerometer signal with a filtered bandwidth of 10 Hz up to a vibration frequency of 500 Hz, and a 100 Hz bandwidth above 500 Hz. The test item shall be energized during vibration and an appropriate input signal shall be applied to observe any abnormal conditions of the output functional characteristics. All failures occurring during the test shall require corrective action before the test is resumed.

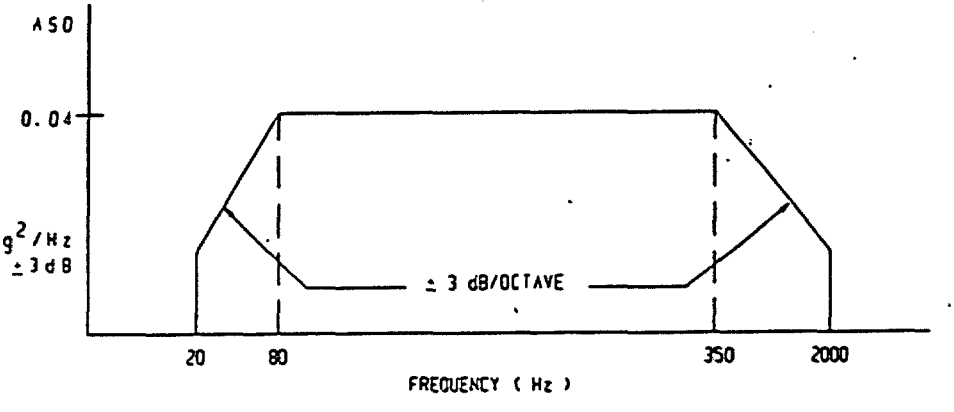


FIGURE 6. Random vibration.

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4.6.2 Temperature screen. Each antenna group shall be subjected to 10 cycles of the temperature curve as specified in FIGURE 7. The temperature rate of change shall be not less than 5°C per minute. Antenna group power shall be turned on and off at the indicated times. Performance measurements on the antenna unit and amplifier-filter unit shall be carried out at the above deck temperatures specified in FIGURE 7. Performance measurements on the switching unit and antenna control unit shall be carried out at the below deck temperatures specified in FIGURE 7. The antenna group shall be positioned for maximum exposure to the changing temperature. Where performance measurements are called for, a minimal functional operating test shall be performed. The dwell time shall be maintained until the largest electrical or electronic part in the antenna group reaches 80 percent of the chamber temperature. When failures occur, the antenna group shall be reworked and the cycling shall be continued for a cumulative total of 10 cycles.

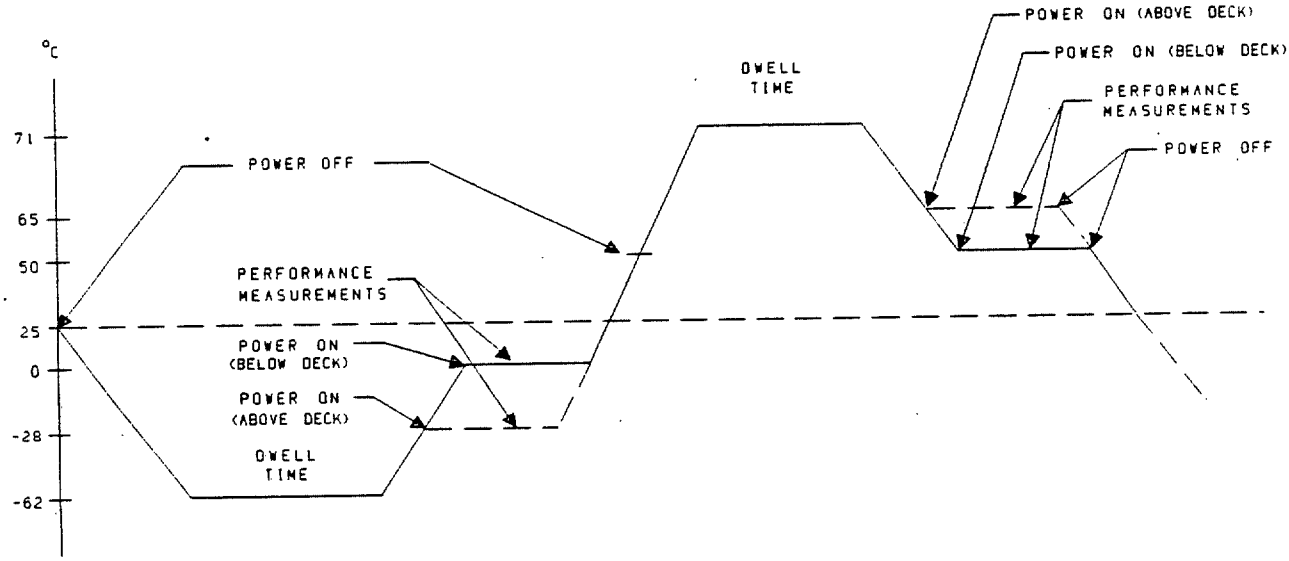


FIGURE 7. One cycle of temperature curve (not to scale).

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4.7 Reliability. Conformance to the reliability requirements of 3.3 shall be verified by the demonstration specified in 4.7.1 through 4.7.3.

4.7.1 Reliability demonstration. A production reliability acceptance test shall be performed on one of the first five production antenna groups in accordance with Task 302 of MIL-STD-781 and MIL-HDBK-781 for category 3A equipment. Testing shall proceed to a decision in accordance with criteria specified for category 3A equipment, test plan VIIID of MIL-HDBK-781. This testing shall be completed to an accept decision before any production antenna groups can be accepted.

4.7.2 Relevant failure. The final decision as to the relevancy or non-relevancy of a failure shall be made by the procuring activity using the Failure categories criteria of MIL-STD-781 as a guide.

4.7.3 Rejected antenna groups. If an antenna group has been rejected as a result of a reliability test, the whole lot from which it was drawn shall be reworked as necessary, reinspected, and another random sample drawn for another reliability test in accordance with 4.7.1. In the event that the cause of failure is antenna group manufacturer design or part manufacturer designs, corrective action shall be performed on all antenna groups delivered under the contract.

4.8 Maintainability. The equipment maintainability requirements shall be demonstrated in accordance with MIL-STD-471 (accept or reject criteria, Method 4) by replacement of subassemblies (modules, individual printed circuit cards) and chassis-mounted electronic, electrical, electromechanical, and mechanical components or parts at the organizational level. The contractor shall determine 60 candidate faults for the demonstration. A representative of the procuring activity shall select a sample of 20 faults from these candidate faults to be used in this part of the demonstration. For fault simulation, a number shall be assigned to each component or part within a subassembly or chassis-mounted component or part. Fault simulation for modules shall be accomplished by introduction of faulty parts corresponding to the number selected.

4.8.1 Demonstration environment. The demonstration at the organizational level shall be performed in a space and temperature environment similar to that in which the equipment will be installed.

4.8.2 Technicians. The procuring activity reserves the right to provide and select the technicians to perform the equipment maintainability demonstration.

4.8.3 Technical documentation. Technical documentation to be used in the demonstration shall be limited to the technical manual.

4.8.4 Rejection. Failure to conform to any of these requirements shall be cause for rejection of the demonstration. If a reject decision is reached, the procuring activity shall be immediately notified.

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4.8.5 Acceptance. The maintainability demonstration shall exhibit that the equipment has conformed to the accept criteria prior to the delivery of any equipment.

4.9 Interchangeability test. The contractor shall perform interchangeability tests to verify conformance with the requirements of 3.1.1.6. Contractor's assemblies shall be substituted into a Government system and proper system operation shall be demonstrated. Government assemblies shall be substituted into the contractor's system and proper system operation shall be demonstrated. Proper system operation shall be demonstrated irregardless of which or how many assemblies are substituted in both tests. These tests shall be performed prior to the delivery of the first production unit and shall be performed at the contractor's plant.

4.10 Inspection of preparation for delivery. Inspections shall be conducted to ensure compliance to the requirements of Section 5.

5. PACKAGING

(The preparation for delivery requirements specified herein apply only for direct Government procurements. Preparation for delivery requirements of referenced documents listed in Section 2 do not apply unless specifically stated in the contract or order. Preparation for delivery requirements for products procured by contractors shall be specified in the individual order.)

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified, preparation for delivery shall be in accordance with the applicable levels of preservation, packaging, and packing specified in MIL-E-17555 (see 6.2).

6. NOTES

6.1 Intended use. The antenna group described herein is intended for shipboard use.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Number of first article samples, if other than specified (see 4.3)
c. If sample antenna groups which have been subjected to and passed production control and environmental inspections may be accepted on the contract (see 4.4.5)
d. Levels of preservation, packaging, packing, and marking (see 5.1)

Preparing Activity:
NAVY - EC

(Project 5985-N564(EC))

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APPENDIX

DERATING CRITERIA FOR ELECTRICAL AND ELECTRONIC PARTS

10. SCOPE

10.1 Scope. This appendix covers the derating criteria for electrical and electronic parts.

20. REQUIREMENTS

20.1 Unless otherwise approved by the Navy, these derating criteria apply to the fabrication during all acquisition phases.

20.2 Derating is increasing the ratio (margin of safety) between part design limits and the allowable applied stresses (electrical, thermal, mechanical, and other environmental stresses). Derating can be accomplished by either selecting a part with broader limits or by limiting the applied stresses. Derating provides added protection for part variances and decreases the part degradation rate thus increasing the expected life of a part.

20.3 The following criteria provide basic information for the derating of electrical and electronic parts and indicate the maximum allowable application stress values to be considered during console design. Since the safety margins of a part to both the threshold and time dependent stresses are generally unknown, parts should be derated to the maximum extent possible consistent with good design practices.

20.4 When derating, the designer shall consider the part manufacturer's specified environmental and operating condition rating versus the actual environmental and operating condition rating of the application and then apply derating factors so as not to exceed the maximum allowable application stresses herein.

Example: A wirewound resistor, RW67U--, which is rated 6.5 watts at 25°C, linearly derating to zero watts at 275°C, being operated in an equipment ambient of 70°C.

a. Compute rated power at 70°C:

$$P_{70\ \text{rated}} = P_{25\ \text{rated}} \times \frac{T_{max} - T_{operating}}{T_{max} - T_{rating}}$$

where $T_{operating} = 70$ and $T_{rating} = 25$

$$= 6.5 \times \frac{275 - 70}{275 - 25} = 6.5 \times \frac{205}{250}$$

$$= 5.33 \text{ watts}$$

b. Apply derating criteria: from tables - limiting power to 0.50 Max P = 5.33 x 50 percent - 2.66 watts

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Part type	Critical stress	Maximum allowable application stress	Remarks
Microcircuits			
Digital	Output current	0.80	
	Operating frequency	0.75	
Linear	Bias voltage	0.70	
	Input signal voltage	0.70	
	Output current	0.75	
	Operating frequency	0.75	
Voltage	Input current (rated maximum)	0.80	
	Output current (rated maximum)	0.75	
	Power dissipation (rated maximum)	0.60	
All	Junction temperature	110°C Max	
All	Ambient temperature	85°C Max	
Transistors	See 1/		
General purpose	Power	0.50	
	Current	0.75	
	Voltage	0.75	
Power	Power	0.30	
	Current	0.75	
	Voltage	0.75	
Switching	Power	0.50	
	Current	0.75	
	Voltage	0.75	
All	Junction temp	110°C Max	
	Leakage current	See remarks	Consider Leakage current to be at least twice rated value.
	Frequency	2/	

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APPENDIX

Part type	Critical stress	Maximum allowable application stress	Remarks
Diodes			
General purpose	Power	0.50	
	Peak inverse voltage (PIV)	0.50	
	Surge current	0.50	
	FWD current	0.50	
Rectifier	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD current	0.50	
Switching	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD current	0.50	
SCR	Power	0.30	
	PIV	0.50	
	Surge current	0.50	
	FWD current	0.50	
Varactor	Power	0.50	
	PIV	0.75	
	FWD current	0.75	
Zener	Power	0.50	
	PIV	0.50	
	Zener current	$0.5\ I_{z_{max}} + I_{z_{nom}}$	
Reference	Power	0.30	
	PIV	0.50	
	Zener current	$0.5\ I_{z_{max}} + I_{z_{nom}}$	
All	Junction temp	110°C MAX	

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APPENDIX

Part type	Critical stress	Maximum allowable application stress 3/	Remarks
Capacitors, Fixed	See 4/		
Ceramic	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Mica, dipped	Voltage	0.60	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Glass	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Porcelain	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Paper	Voltage	0.50	Good low frequency characteristics
	Current		
	AC/ripple	0.70	
Surge		0.70	
Plastic	Voltage	0.50	Mylar good to 1 MHz; teflon polystyrene, and polycarbonate good to 1 GHz
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Tantalum, foil or film	Voltage	0.50	Limit to 70°C ambient temperature
	Current		
	AC/ripple	0.70	
	Surge	0.70	

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Part type	Critical stress	Maximum allowable application stress 3/	Remarks
Tantalum, solid, dry	Voltage	0.50	Limit to 85°C ambient temperature
	Current		
	AC/ripple	0.70	
	Surge protection	3 ohms/volt	
Tantalum solid, wet	Voltage	See remarks	Use of tantalum wet slug capacitors is prohibited except by prior approval of the procuring activity. 5/
	Current		
	AC/ripple	See remarks	
Capacitors adjustable			
Ceramic	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Glass	Voltage	0.50	
	Current		
	AC/ripple	0.70	
	Surge	0.70	
Air	Voltage	0.30	
All, fixed & adjustable	Leakage	See remarks	For all applications, consider leakage current as twice rated maximum value

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Part type	Critical stress	Maximum allowable application stress	Remarks
Resistors, fixed	See 6/		
Carbon composition	Power	0.50	
Insulated film	Power	0.50	
Wirewound precision			
1.0 percent total	Power	0.50	
0.1 percent total	Power	0.25	
Wirewound power	Power	0.50	
Thermistor	Power	0.50	Uses in other than zero power applications shall specify minimum wattage for the application.
Adjustable	See 6/		
Wirewound	Rated current	0.70	See 7/
Non-Wirewound	Rated current	0.70	See 7/
All	Voltage	0.80	Maximum voltage shall not exceed 80 percent of the MIL-Ratings.

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Part type	Critical stress	Maximum allowable application stress	Remarks
Relays and switches load type			
Capacitive	Contact current	0.75	Peak inrush current shall not exceed derated limit.
Resistive	Contact current	0.75	
Inductive	Contact current	0.40	
Motor	Contact current	0.20	
Filament	Contact current	0.10	
Miscellaneous			
Transformer and chokes	Flux saturation level Insulation breakdown voltage Current	0.75 0.50 0.70	Parts shall be at winding temperature at least 30°C below rating. Maximum voltage between winding and shield shall not exceed 60 percent of rating for steady-state conditions or 90 percent of rating with transients.

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Part type	Critical stress	Maximum allowable application stress	Remarks
Connectors	Current, pin	See remarks	Current de-rating for connector pins shall be in accordance with criteria for relays and switches, depending on type of loading. For paralleled pins sum of current must not be more than 100 percent rated of lowest rated pin.
	Insulation breakdown voltage	0.50	

1/ Worst-case combination of DC, AC, and transient voltages shall be no greater than derated limit.

2/ Transistors rated at 500 milliwatts or less in TO-5 or TO-12 packages using one MIL or smaller aluminum lead wires with thermocomposition bonds shall not be used at repetition rates of less than 30 Hz. At 30 Hz and 25°C or less, derate the power dissipation to 0.20 and limit the input current to 50 mA. Above 25°C, derate an additional 1.33 mW/°C. Gold lead devices should be used at repetition rates less than 30 Hz.

3/ Manufacturer's derating factors shall be applied before applying these factors.

4/ Voltage equals instantaneous total of DC, AC, surge, and transient voltages.

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5/ When approved for use, tantalum wet slug capacitors shall be subjected to special screens such as temperature cycling and seal leakage.

6/ High density packaging may require further derating if ambient temperatures are increased.

7/ Rated current is defined as:

$$I_R = \frac{P_{max}}{R_{max}}$$

And by limiting the current to 0.70 rated current, power is limited to 0.50 maximum power.

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